Supplemental Material: Experimental Evaluation of High-Throughput Structure Predictions

S1. SAMPLES

Table S1 lists the 59 arc-melted buttons made for the 34 candidate compositions, with the mass loss of each. For each candidate, one pattern, the pattern of record, gives the outcome in Table I, and its fit is the fit of record. Table S1 gives the button of each pattern, whether it was as cast or annealed, the instrument and the mount. For four candidates, YAl_3Cu_2, $ScCrC_2$, Sc_2Pd_5Au and Pd_6Cu_5Ge, this pattern is of a later button of the same composition. Nine samples were annealed (Methods). For eight of them, the outcome comes from the annealed pattern, and for Sc_2PdCu, whose annealed pattern was inconclusive, from the as-cast pattern. Section S7 shows, for each candidate, the fit of its pattern and the reasoning behind its outcome in Table I.

S2. PHASE IDENTIFICATION

We first analyzed every pattern with MDI JADE, used interactively, by search-match against the PDF-5+ 2026 database of the International Centre for Diffraction Data (ICDD) [1] and by whole-pattern fitting. We also searched the predicted structures of OQMD, MP and Alexandria through the JADE user database, but not in the same search as the PDF cards, and without disordered versions of these structures. This gave a first outcome for each sample and in some cases identified the main phases, but several patterns were left only partly explained.

Following this manual analysis, we built an automated pipeline that matches each pattern against all of these structures together. For each ternary system, the library holds the elemental, binary and ternary entries of OQMD, MP and Alexandria, the structures of the system in the Crystallography Open Database (COD) [2], and the PDF-5+ cards used in the JADE analysis. Because a disordered phase can give the same main lines as an ordered predicted structure, the library also holds a fully disordered version of each ordered structure, in which every site carries the average composition. For many systems it also holds partially disordered versions, in which only some sites mix, with cells relaxed with MatterSim, and solid solutions between two known binaries of the same structure type that share an element, such as (Pt,Ni)Ge or Sc(Pd,Cu), with lattice parameters interpolated by Vegard's law. It also holds random fcc, bcc and hcp cells at the nominal composition, whose volume per atom is the composition-weighted average of the volumes per atom of the elements. The calculated pattern of each phase includes the instrumental line profile. Because computed lattice parameters differ slightly from measured ones, the search recalculates the pattern of each phase with its lattice parameters scaled by up to 2% and keeps the scaling that best matches the measured pattern. The candidates are single phases, pairs and triples whose compositions combine to give the nominal composition to within 0.15 in barycentric composition coordinates, and pairs of polymorphs at one composition. Each combination is fitted to the pattern by non-negative least squares and scored by its residual, its energy above the hull and small penalties for extra phases, trace phases and an off-nominal composition, with weights set in the deposited code. The 15 best-ranked combinations, chosen to differ in their phases, were refined in GSAS-II [3] and ranked by R_{wp}, and combinations whose refined parameters became unphysical were discarded. The repository [4] contains the search-match code but not the PDF-5+ cards, which may not be redistributed, and the final Rietveld fits start only from open structures (Sec. S3).

The automated search did not reliably find phases below about 10 wt%. We added such phases by hand, either from the JADE analysis or by testing likely extra phases, such as oxides, against lines the fit left unexplained, and we kept a phase only where lines of its own, not shared with another phase, are present in the data. With that check, the final fits determine smaller fractions, such as the 6 wt% of Gd_2O_3 in $GdPt_2Ge$.

S3. X-RAY DIFFRACTION AND RIETVELD REFINEMENT

All patterns were measured with Cu Kα radiation, all but two on the Siemens D500. Powder ground or filed from the button was mounted on a PMMA slide or, for most annealed samples, on a zero-background Si slide. On PMMA slides the powder was fixed with silicone grease or petroleum jelly, and the slide and the grease add a broad hump at low angle, so we started each PMMA-slide refinement above it, mostly near 18°. On zero-background slides a few drops of grease diluted to a few percent in hexane left a thin film, onto which the powder was sprinkled, so these patterns have almost no low-angle hump. The D500 scans run from 5 to 120 or 150° in steps of 0.02°, and the two PANalytical scans from about 20 to 120 and 30 to 130°. Three patterns are of solid pieces of a button rather than powders, two of them measured on a PANalytical X'Pert MPD, and their entries in Sec. S7 describe each.

For the D500, we refined the instrumental line profile and zero offset in GSAS-II [3] from a pattern of a quartz standard and held them in every final fit. Only the as-cast fits of Sec. S4 refine a zero offset, in place of the sample displacement. The PANalytical patterns were fitted with that instrument's profile from a LaB_6

TABLE S1: Arc-melted buttons of the 34 candidates, grouped and numbered as in Table I. Button: the button whose sample number identifies the candidate in the paper. Earlier attempt: a button of the same composition that this button replaced. Remake: a later button of the same composition. Mass loss in percent in parentheses, with – where it was not recorded. †: not scanned. Pattern of record: the pattern that gives the outcome, as cast or annealed, instrument (D500, Siemens D500; PAN, PANalytical), mount (ZB, powder on a zero-background Si slide; PMMA, powder on a poly(methyl methacrylate) slide; bulk, a piece of the button, measured without grinding). Annealed: 1000 °C for one week.

#	Compound	Button	Earlier attempt	Remake	Pattern of record	
On-hull structure						
1	$ZrPt_5Si$	0561 (1.0)	0482 (0.3)		0561	annealed, D500, ZB
2	YAl_3Cu_2	0540 (3.0)	0506 (14.3)	0615 (2.1)	0615	annealed, D500, ZB
3	$GdPt_2Ge$	0480 (0.3)			0480	annealed, D500, ZB
4	$TiPt_5Si$	0493 (1.9)			0493	annealed, D500, ZB
5	YSnCu	0507 (0.4)			0507	as cast, D500, PMMA
6	$ScCrC_2$	0512 (3.0)		0605 (–)	0605	as cast, D500, PMMA
Above-hull polymorph						
7	YCoSi	0538 (<0.1)	0503 (<0.1)		0538	annealed, D500, ZB
8	$YCoSi_3$	0539 (0.5)	0504 (2.9)		0539	annealed, D500, ZB
Disordered parent of on-hull structure						
9	Al_8FeCo_7	0535 (0.6)	0487 (9.4)		0535	as cast, D500, PMMA
10	Hf_2MoIr	0466 (0.6)			0466	annealed, D500, ZB
11	Al_4FeCo_3	0545 (0.8)	0486 (3.9)	0534 (3.3), 0544 (10.9)†	0545	as cast, D500, PMMA
12	Sc_2PdCu	0474 (0.1)			0474	as cast, PAN, bulk
13	Sc_3Pd_2Cu	0473 (0.2)			0473	annealed, D500, PMMA
14	$ScPd_5Au_2$	0477 (15.6)			0477	as cast, PAN, bulk
15	Al_2MnAu_6	0479 (2.0)			0479	as cast, D500, PMMA
16	Sc_2Pd_5Au	0478 (0.5)		0610 (1.0)	0610	as cast, D500, PMMA
17	$PtNiGe_2$	0481 (0.2)			0481	as cast, D500, PMMA
18	Gd_3Sn_6Co	0556 (2.0)	0489 (4.5)	0536 (9.5)	0556	as cast, D500, PMMA
Other solid solution						
19	$ScPd_2Au$	0476 (0.2)			0476	as cast, D500, PMMA
Multiphase, on-hull structure not formed						
20	$ScPd_2Cu$	0469 (1.2)			0469	as cast, D500, PMMA
21	$La_5Al_3Co_2$	0483 (<0.1)			0483	as cast, D500, PMMA
22	$TiPt_5Ge$	0491 (1.4)			0491	as cast, D500, PMMA
23	Ti_2Pt_8Si	0494 (0.4)			0494	as cast, D500, PMMA
24	$LaNiSi_3$	0514 (2.2)			0514	as cast, D500, PMMA
25	$GdSn_6Co_4$	0537 (0.5)	0490 (11.1)		0537	as cast, D500, PMMA
26	$NbNi_2B_2$	0510 (0.5)			0510	as cast, D500, PMMA
27	$GdFeGe_2$	0555 (1.5)	0497 (5.4)		0555	as cast, D500, PMMA
28	$Gd_2Fe_3Ge_5$	0554 (4.5)	0496 (1.8)	0557 (5.0)†	0554	as cast, D500, PMMA
29	$GdFeGe_3$	0553 (2.3)	0495 (2.3)		0553	as cast, D500, PMMA
30	$LaRuC_2$	0513 (6.3)		0614 (0.6), 0617 (2.2)	0614	as cast, D500, PMMA
On-hull structure did not form but structure(s) unresolved						
31	Pd_6Cu_5Ge	0465 (0.3)		0568 (1.6), 0608 (1.1)	0568	as cast, D500, PMMA
32	La_2PdAu_2	0475 (0.9)			0475	as cast, D500, PMMA
33	Ti_2Pt_8Ge	0492 (0.5)			0492	as cast, D500, bulk
34	$ScPdCu_4$	0472 (5.1)		0558 (13.0)†, 0566 (4.7), 0613 (4.5)	0472	as cast, D500, PMMA

standard.

With the zero offset held, we refined the sample displacement, the height of the powder surface relative to the axis of the diffractometer. For seven powders on PMMA slides, noted in their entries, it refined to more than about 300 µm. A thin layer of powder on a flat slide cannot be that far out of position, so we held it at zero for these seven. Any real displacement then shifts their lines without being modelled, so the lattice parameters of these seven samples are uncertain by up to about 0.5%. For thin layers of powder on zero-background slides, noted in their entries, we refined a thin-layer intensity correction, which agrees with the mass of powder on the slide.

A script ran each refinement by adding one group of parameters at a time. The scale factors with the background, the phase fractions, the main-phase lattice parameters with the sample displacement, the preferred-orientation term and the order parameter were kept whenever their values were physical. Every other group was kept only if it lowered R_{wp} (defined below) by more than 0.05 percentage points and left every value physi-

cal, which means positive microstrain and crystallite size, occupancies between 0 and 1 and a March–Dollase ratio between 0.3 and 3. In order, it refined the scale factors and a Chebyshev background, the phase fractions, and the lattice parameters of the main phase with the sample displacement. It then refined the lattice parameters of each other phase, the microstrain and crystallite size of the main phase, and the microstrain of the larger other phases. Where the pattern required them, it added an anisotropic microstrain [5], a March–Dollase preferred-orientation term [6], and site occupancies or the single order parameter described below. It ended with an isotropic atomic displacement parameter for the main phase.

Each lattice parameter could move at most 2% from its starting value for a structure taken from a database or the literature, and at most 5% for a disordered parent or a solid solution, whose starting cell is only an estimate. The script undid any step that moved a lattice parameter beyond this limit. A predicted structure whose lines cannot be matched within 2% of its computed lattice parameters therefore counts as absent.

R_{wp} is the usual weighted profile residual [7]. Because R_{wp} depends on the range, the counting statistics and the background, we compare it only between fits of the same scan over the same range with the same options.

The final fits start only from open structures: entries of OQMD, MP, Alexandria and the COD, cells we built for disordered and solid-solution models, and six structures we entered by hand from the crystallographic tables of their publications, so that the repository holds no copyrighted PDF cards. Two are $CeNiSi_2$-type phases, $GdFe_{0.37}Ge_2$ of Jemmali et al. [8] and $GdCo_xSn_2$ of Gil et al. [9]. The others are $ScCrC_2$ of Pöttgen et al. [10], $LaNi_6Si_6$ of Pani et al. [11], $La_4Co_3Al_3$ of Tursina et al. [12], and the La_3Al_{11}-type $Y_3(Al,Cu)_{11}$ partner of YAl_3Cu_2 from the thesis of Krachan [13]. The disordered parents and the models of partial order were built from the predicted structures. In a model of partial order, one parameter moves the occupancy of each site linearly from its value in the ordered structure to the average over the sites that mix, that is, from full order to a random distribution.

A candidate counts as formed when the phase at the nominal composition is the largest phase by weight fraction in its fit. To test for order, we fitted the pattern with the ordered on-hull structure and with its disordered parent, the same structure with only the elements that the order separates mixed on their sites. At each line that the order adds or changes, we compared the net observed intensity, after subtracting the background, with the calculated intensity, using the counting uncertainty of the observed intensity. If the ordered model calculates these lines above that uncertainty and they are observed, the order is present. If they are calculated above the uncertainty but not observed, the ordered structure is absent and the candidate is a disordered parent. If they are calculated below the uncertainty, as for the Fe/Co order of Al_8FeCo_7 and Al_4FeCo_3, the two cannot be told apart, and Fig. 2 marks both. Section S7 shows, for each candidate, the fit of its pattern together with one or two fits of alternative models of the same pattern, such as the predicted ordered structure in place of the phase that formed, or a model in which the ordered elements share their sites. The fitted patterns, refinement logs and GSAS-II project files of every fit in Sec. S7 are in the repository [4].

S4. AS-CAST AND ANNEALED SAMPLES

Figure S1 compares the fits before and after annealing for the three new compounds. $GdPt_2Ge$ was already the main phase as cast, and the *Pnma* phase alone fits the as-cast pattern once a preferred orientation along [100] is included. $TiPt_5Si$ is 62 wt% of the as-cast button, and its (100) superlattice line near 22.7° is clear of the lines of $TiPt_8$ and has the calculated intensity. Annealing raised the $TiPt_5Si$ fraction to 91 wt%, and that of $ZrPt_5Si$ from 29 to 97 wt%. The as-cast fits have a lower R_{wp} than the annealed ones, although their patterns are poorer. The as-cast powders were on PMMA slides and were refined over a shorter range, and the slide background makes up most of their counts. Because R_{wp} includes the background, it is smaller for such patterns, so it cannot be compared between the two scans of a sample.

S5. DATABASE VALUES

Table S2 gives $\Delta H_{\mathrm{f}} - C(x)$ for each candidate in the three databases, computed as described in Methods. The values were computed from the entries of each ternary system in OQMD 1.7, the Materials Project (April 2026) and the Alexandria PBE release 2025.07.02. These entries, extracted from the full databases, are included in the repository [4].

S6. DYNAMICAL STABILITY

Table S3 lists, for each candidate, the structures that lie below the hull of the other compositions in at least one database, and whether each is dynamically stable. We relaxed each structure with MatterSim 1.2.5 (5M model) and calculated its phonons with phonopy 4.8.0 by finite displacements in a $2 \times 2 \times 2$ supercell. A structure counts as unstable if any frequency on the seekpath band path lies below -0.1 THz. Entries of different databases count as one structure when they have the same space group and pymatgen's StructureMatcher matches them.

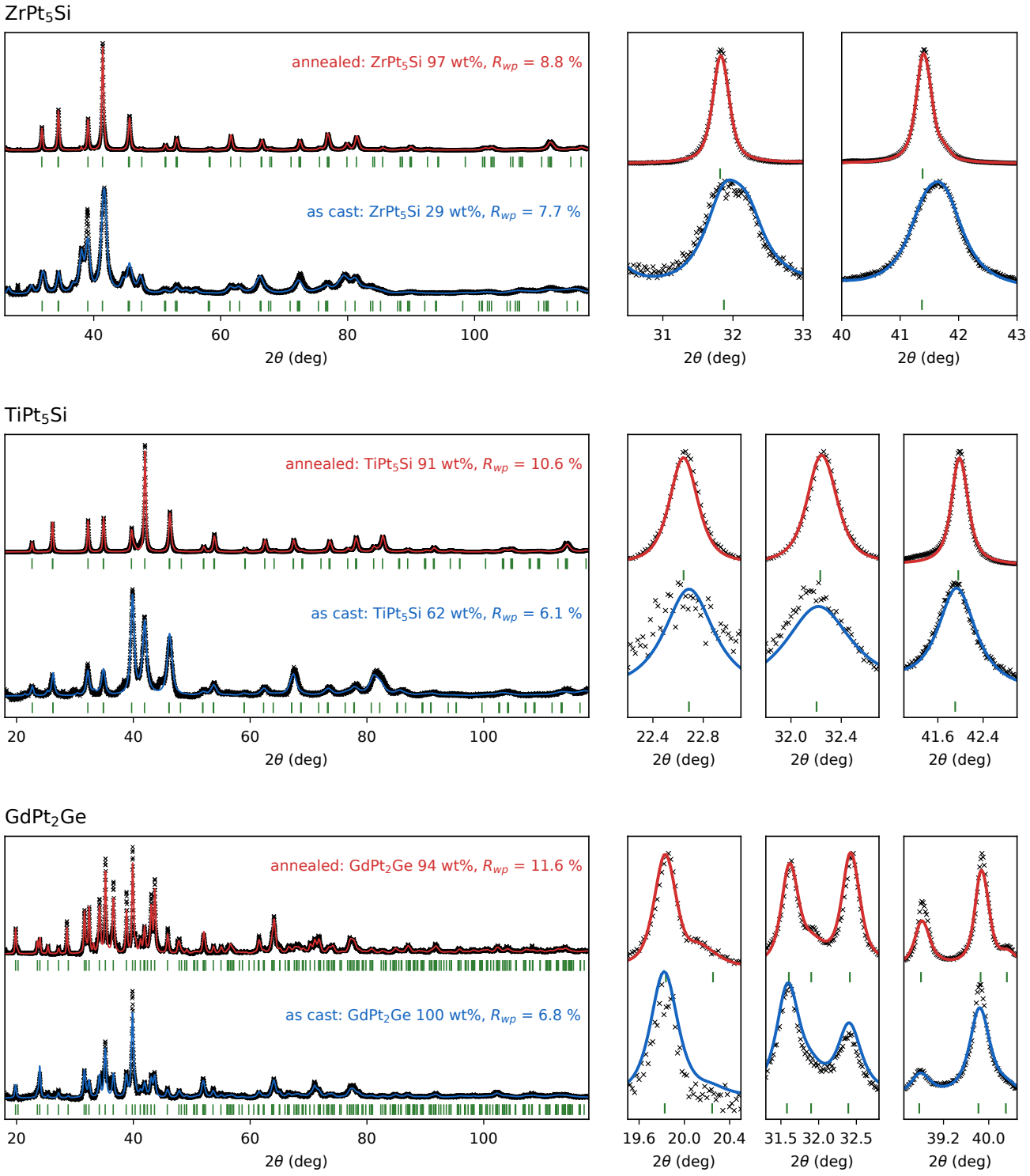


FIG. S1: Fits of the three new compounds after annealing and as cast. Here the fit after annealing is red and the as-cast fit blue (no difference curves are drawn). Left, full patterns, with the measured pattern minus the background (crosses) and the Bragg positions of the phase at the nominal composition (green ticks), labeled with its weight fraction and R_{wp}. Right, lines of that phase enlarged. R_{wp} cannot be compared between the as-cast and annealed scans (Sec. S4). Annealing raised the fractions of $TiPt_5Si$ and $ZrPt_5Si$, while $GdPt_2Ge$ was already the main phase as cast.

TABLE S2: The 34 candidates as seen by the three databases. For each database, $\Delta H_\mathrm{f} - C(x)$ is the energy of its lowest entry at the nominal composition relative to the convex hull of all other compositions in that database (meV/atom; negative means the database predicts the compound to be stable), recomputed from the cached entries by the same construction for all three. "Same" states whether the database's ground-state structure at that composition is the same structure as the OQMD prediction from which the candidate was selected (pymatgen StructureMatcher); the space group is given where it differs. Blank: no entry at the composition. Grouped and numbered as in Table I. Snapshots: OQMD v1.7 local dump; Materials Project API, 7 April 2026; Alexandria PBE 2025.07.02 release.

		OQMD		Materials Project		Alexandria (PBE)	
#	Formula	$\Delta H_\mathrm{f} - C$	Structure	$\Delta H_\mathrm{f} - C$	Same	$\Delta H_\mathrm{f} - C$	Same
On-hull structure							
1	$ZrPt_5Si$	-41.8	$P4/mmm$	-41.8	yes	-21.5	yes
2	YAl_3Cu_2	-10.8	$P6/mmm$			-3.0	yes
3	$GdPt_2Ge$	-32.5	$Cmcm$			-25.4	no ($Pnma$)
4	$TiPt_5Si$	-16.5	$P4/mmm$			-25.6	yes
5	YSnCu	-69.1	$P6_3mc$	-69.0	yes	-69.6	yes
6	$ScCrC_2$	-10.0	$P6_3/mmc$	26.1	yes	26.8	yes
Above-hull polymorph							
7	YCoSi	-42.9	$P4/nmm$	16.9	no ($Pnma$)	-31.1	yes
8	$YCoSi_3$	-17.3	$I4mm$			-14.9	yes
Disordered parent of on-hull structure							
9	Al_8FeCo_7	-3.4	$Pm\bar{3}m$	12.1	yes		
10	Hf_2MoIr	-12.0	$Fm\bar{3}m$	-9.1	yes	3.7	yes
11	Al_4FeCo_3	-6.5	$Im\bar{3}m$			-3.2	no ($R\bar{3}m$)
12	Sc_2PdCu	4.8	$Fm\bar{3}m$	-1.3	yes	-0.7	no ($P4/mmm$)
13	Sc_3Pd_2Cu	-3.0	$P4/mmm$			4.9	yes
14	$ScPd_5Au_2$	2.8	$Pbam$			9.9	yes
15	Al_2MnAu_6	-0.6	$C2/m$			3.2	yes
16	Sc_2Pd_5Au	-9.6	$Pbam$	-12.9	no ($Amm2$)	-34.7	yes
17	$PtNiGe_2$	-5.8	$P2_1/m$			17.9	no ($P\bar{3}m1$)
18	Gd_3Sn_6Co	-25.1	$Cmcm$				
Other solid solution							
19	$ScPd_2Au$	6.4	$P4/mmm$	2078.0	no ($Immm$)	-4.7	yes
Multiphase, on-hull structure not formed							
20	$ScPd_2Cu$	17.7	$I4/mmm$	33.2	no ($Fm\bar{3}m$)	4.8	no ($P4/mmm$)
21	$La_5Al_3Co_2$	-0.7	$Cmcm$				
22	$TiPt_5Ge$	-18.8	$P4/mmm$			-34.0	yes
23	Ti_2Pt_8Si	-4.8	$P4/mmm$			90.4	no ($Immm$)
24	$LaNiSi_3$	-24.7	$I4mm$			-5.2	yes
25	$GdSn_6Co_4$	-4.1	$I\bar{4}2m$			-11.2	yes
26	$NbNi_2B_2$	12.1	$I4/mmm$			23.0	yes
27	$GdFeGe_2$	0.2	$P2_1/m$			-6.3	yes
28	$Gd_2Fe_3Ge_5$	-8.6	$C2/c$				
29	$GdFeGe_3$	-19.0	$I4mm$			-30.6	yes
30	$LaRuC_2$	0.9	$Cmcm$			28.3	no ($Amm2$)
On-hull structure did not form but structure(s) unresolved							
31	Pd_6Cu_5Ge	-19.6	$Cmmm$			-15.9	no ($Immm$)
32	La_2PdAu_2	-0.8	$C2/m$			11.9	yes
33	Ti_2Pt_8Ge	-3.9	$P4/mmm$	-45.4	yes	103.0	no ($Immm$)
34	$ScPdCu_4$	-24.2	$F\bar{4}3m$			3.9	yes

TABLE S3: Dynamical stability of the structures below the hull at each candidate composition, grouped and numbered as in Table I. Each structure is given by its space group, with the databases that contain it (O, OQMD; M, MP; A, Alexandria PBE), in order of increasing $\Delta H_\mathrm{f} - C(x)$. A structure is listed if it lies below the hull of the other compositions in at least one of these databases. Stable: no phonon frequency below -0.1 THz on the band path. $\times$: every structure below the hull is dynamically unstable (crosses in Fig. 2).

#	Compound	Dynamically stable	Dynamically unstable	
On-hull structure				
1	$ZrPt_5Si$	$P4/mmm$ (O,M,A)	–	
2	YAl_3Cu_2	$P6/mmm$ (O,A)	–	
3	$GdPt_2Ge$	$Cmcm$ (O,A), $Pnma$ (A)	–	
4	$TiPt_5Si$	$P4/mmm$ (O,A)	–	
5	YSnCu	$P6_3mc$ (O,M,A), $F\bar{4}3m$ (A), $Pnma$ (O,A), $P4/nmm$ (A), $P\bar{6}m2$ (A), $Pnma$ (A), $Imm2$ (A)	$P6_3/mmc$ (O)	
6	$ScCrC_2$	$P6_3/mmc$ (O,M,A), $Pmmn$ (O,M)	–	
Above-hull polymorph				
7	YCoSi	$P4/nmm$ (O,A), $Pnma$ (O,M,A)	–	
8	$YCoSi_3$	$Cmmm$ (O), $P2/m$ (A)	$I4mm$ (O,A)	
Disordered parent of on-hull structure				
9	Al_8FeCo_7	$Pm\bar{3}m$ (O,M)	–	
10	Hf_2MoIr	$Fm\bar{3}m$ (O,M,A), $P4/mmm$ (O,A)	–	
11	Al_4FeCo_3	$Im\bar{3}m$ (O), $R\bar{3}m$ (A), $P4/mmm$ (A)	–	
12	Sc_2PdCu	–	$Fm\bar{3}m$ (O,M,A), $P4/mmm$ (A)	$\times$
13	Sc_3Pd_2Cu	–	$P4/mmm$ (O,A)	$\times$
14	$ScPd_5Au_2$	none below the hull	–	
15	Al_2MnAu_6	–	$C2/m$ (O,A)	$\times$
16	Sc_2Pd_5Au	$Pbam$ (O,A), $I4/mmm$ (A), $Amm2$ (M)	–	
17	$PtNiGe_2$	$P2_1/m$ (O)	–	
18	Gd_3Sn_6Co	$Cmcm$ (O)	–	
Other solid solution				
19	$ScPd_2Au$	$P4/mmm$ (O,A)	–	
Multiphase, on-hull structure not formed				
20	$ScPd_2Cu$	none below the hull	–	
21	$La_5Al_3Co_2$	–	$Cmcm$ (O)	$\times$
22	$TiPt_5Ge$	$P4/mmm$ (O,A)	–	
23	Ti_2Pt_8Si	$P4/mmm$ (O)	–	
24	$LaNiSi_3$	–	$I4mm$ (O,A)	$\times$
25	$GdSn_6Co_4$	–	$I\bar{4}2m$ (O,A)	$\times$
26	$NbNi_2B_2$	none below the hull	–	
27	$GdFeGe_2$	$P2_1/m$ (O,A)	–	
28	$Gd_2Fe_3Ge_5$	$C2/c$ (O), $Ibam$ (O)	–	
29	$GdFeGe_3$	–	$I4mm$ (O,A)	$\times$
30	$LaRuC_2$	none below the hull	–	
On-hull structure did not form but structure(s) unresolved				
31	Pd_6Cu_5Ge	–	$Cmmm$ (O,A), $Immm$ (A)	$\times$
32	La_2PdAu_2	$C2/m$ (O,A)	–	
33	Ti_2Pt_8Ge	$P4/mmm$ (O,M)	–	
34	$ScPdCu_4$	$F\bar{4}3m$ (O,A)	–	

S7. PER-SAMPLE ANALYSIS

For each candidate, in the order of Table I, this section gives the fit of record, whose phases define the outcome, and one or two fits of alternative models. Each refines the same scan with the same range and options but changes one model, usually by adding the predicted ordered structure, putting it in place of a phase, or mixing its sites. In panel (a), crosses show the measured pattern minus the fitted background, the red line the fit of record and the dark blue line the difference. The rows of tick marks below the pattern, numbered as in the legend, give the Bragg positions of each phase, green for the phase at the nominal composition. Panel (b) compares the difference curves of the fit of record (red) and of the alternative models (blue dashed, then black dotted), each labeled with its R_{wp}. Panel (c) enlarges the deciding lines, marked by gray bands in (a), with each model's calculated pattern in its color, or its stick pattern where its refinement diverged.

S7.1. Samples in which the on-hull structure formed

1. $ZrPt_5Si$

After a week at 1000 °C, $ZrPt_5Si$ is 97 wt% of its sample in the predicted $P4/mmm$ structure, with about 3 wt% of what is likely $ZrPt_3$ (Fig. S2). Lines at 38.0° and 47.4°, with about 2.5% and 1.5% of the intensity of the strongest line, fall at positions of $ZrPt_3$ lines (Fig. S2d). $ZrPt_5Si$ gives no line at either position. With the same adjustable parameters, the fit without $ZrPt_3$ leaves these lines out and is worse (R_{wp} = 10.5% against 8.4%). The measured lines are broader than those calculated for $ZrPt_3$, so the identification is not certain. A model with Zr, Si and Pt mixed on all sites fits far worse (R_{wp} = 23.6% against 8.4%) and calculates the (110) line at 31.8° at one eighth of its observed intensity, so Pt is ordered on its own sites. Allowing Zr and Si to exchange gives an exchange of about 1% and no better fit, so Zr and Si are ordered as well. Every model calculates the (001) line at 12.8° too strong, which Zr/Si exchange does not explain. In the as-cast button, $ZrPt_5Si$ was a minority phase (Sec. S4). The prediction is borne out: $ZrPt_5Si$ formed in the predicted ordered structure.

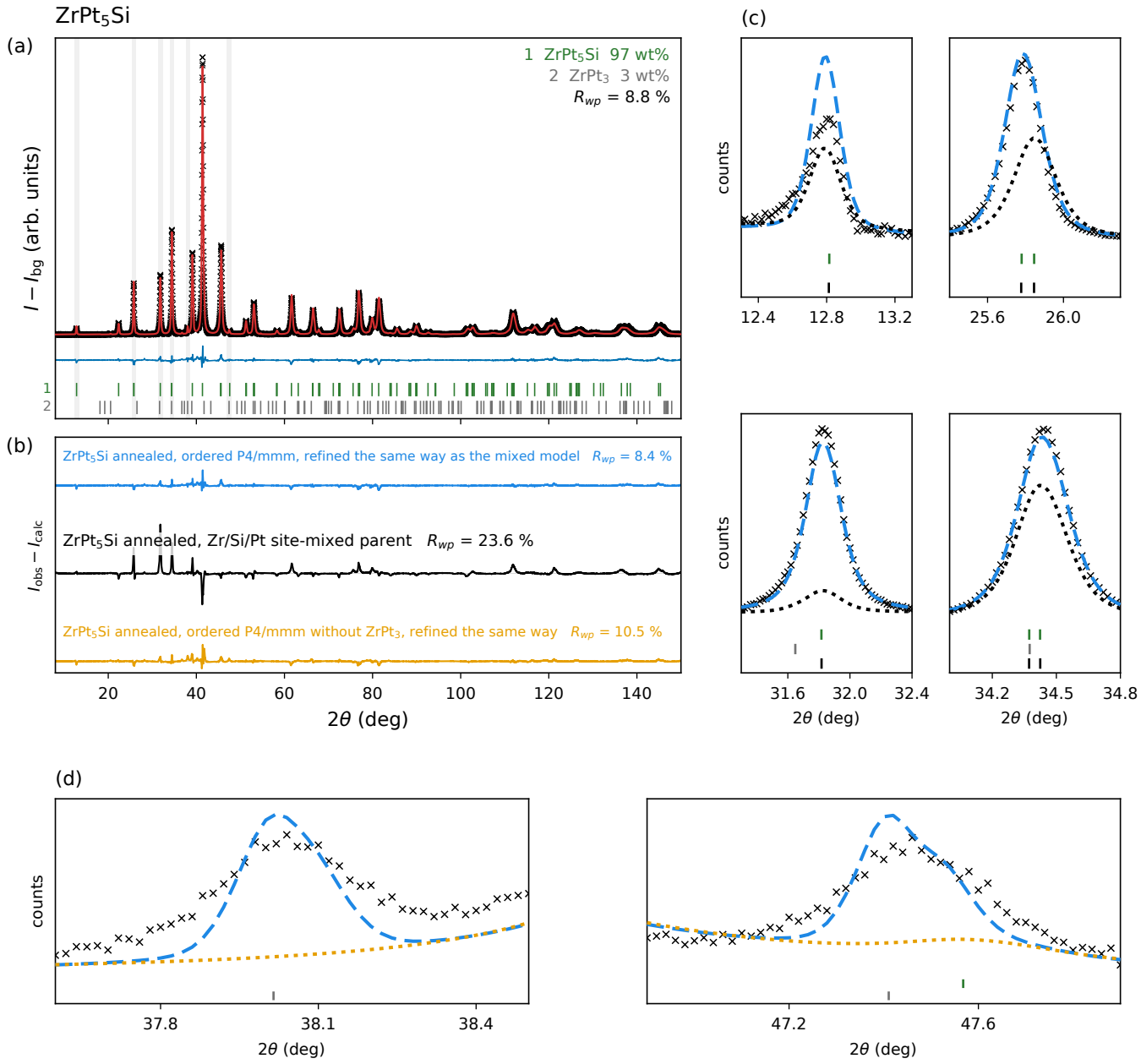


FIG. S2: Annealed $ZrPt_5Si$. Unlike the other figures, (b)–(d) leave out the fit of record and compare three models refined with the same adjustable parameters: the ordered structure with $ZrPt_3$ (blue dashed), the structure with Zr, Si and Pt mixed on all sites, with $ZrPt_3$ (black dotted), and the ordered structure without $ZrPt_3$ (orange dotted). In (c), blue and black differ only in their site occupancies, and the site-mixed model falls far below the (110) line near 31.8°. In (d), only the model with $ZrPt_3$ fits the weak lines near 38.0° and 47.4°.

2. YAl_3Cu_2

The fit of annealed YAl_3Cu_2 is poor ($R_{wp} = 11.9\%$), with large residuals at the main lines and several lines left unfitted (Fig. S3). It holds two phases, a $CaCu_5$-type phase in the predicted structure and probably a La_3Al_{11}-type $Y_3(Al,Cu)_{11}$ phase [13], but lines strongest near 29.1, 35.2 and 41.2° belong to neither, and some main lines are shifted or have shoulders, so the two phases do not describe the whole sample and we quote no weight fractions. The conclusion that the predicted Al/Cu order formed rests on one line that the misfit does not affect. The (001) line of the $CaCu_5$-type phase at 21.7° is clearly present in the data. A model with Al and Cu at random on the 3g and 2c sites gives this line less than a fifth of its observed intensity. The model with the predicted order, Al on 3g and Cu on 2c, gives the line somewhat more intensity than observed, and a partly ordered model, with Al on 88% of the 3g sites and Cu on 83% of the 2c sites, matches the observed intensity. The predicted structure formed, with Al and Cu mostly on their predicted sites, alongside at least two other phases.

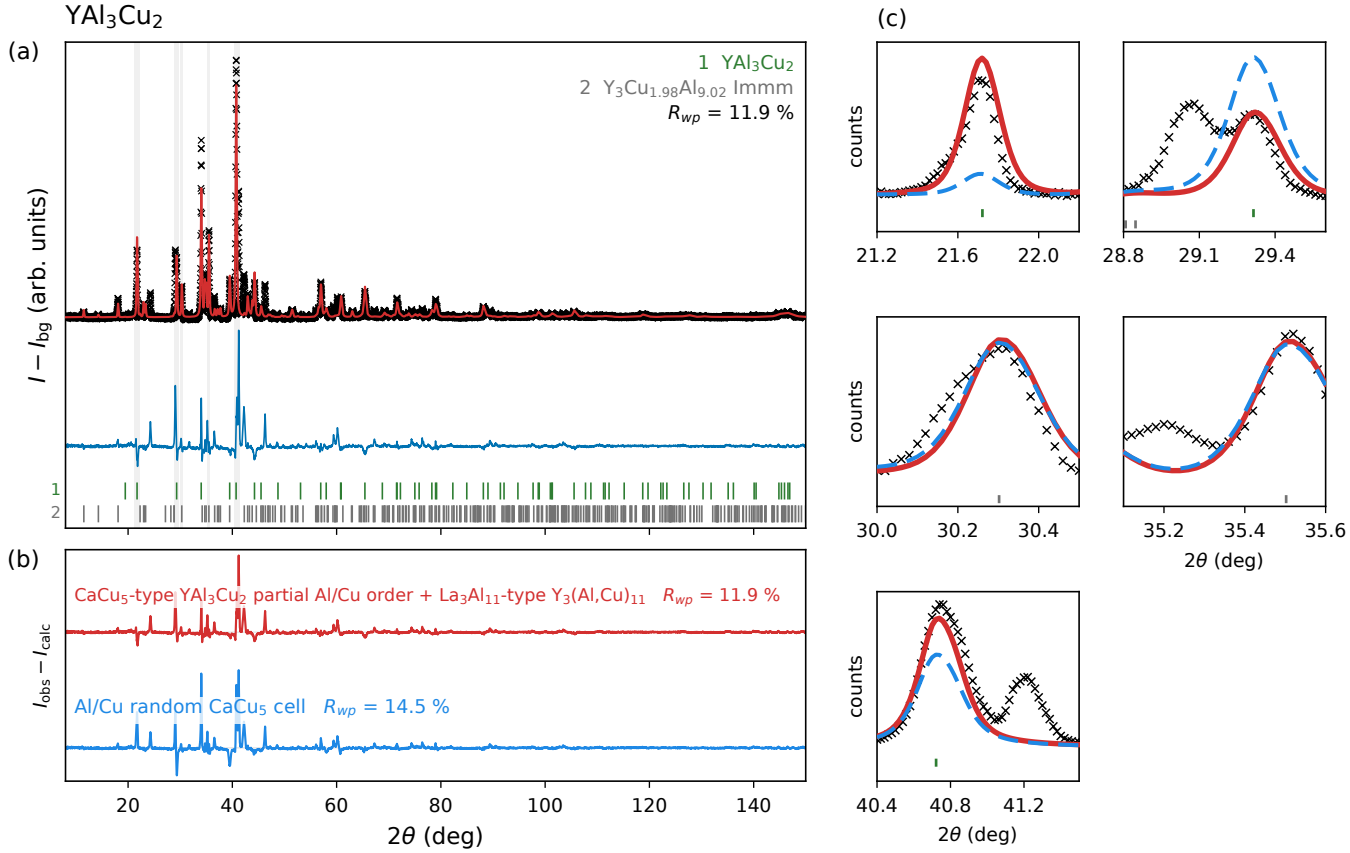


FIG. S3: Annealed YAl_3Cu_2. The alternative model in (b) places Al and Cu at random on the 3g and 2c sites of the same $CaCu_5$-type cell. In (c), only the partly ordered model reproduces the (001) line at 21.7°. The unfitted lines near 29.1°, 35.2° and 41.2° belong to the unidentified phase. Thin-layer intensity correction refined (Sec. S3).

3. $GdPt_2Ge$

After annealing, 94 wt% of the $GdPt_2Ge$ sample has the YPd_2Si-type *Pnma* structure, and the rest is likely Gd_2O_3 (Fig. S4). This *Pnma* structure is the lowest-energy structure at this composition in Alexandria. The *Cmcm* structure, which OQMD lists as the lowest-energy structure, did not form.

The *Cmcm* structure does not fit the pattern at all. Refined alone, it gives $R_{\mathrm{wp}} = 41\%$, against 14% for the *Pnma* structure refined the same way. It gives no intensity at the strong observed lines at 19.8, 31.6, 32.4, 38.8 and 39.9°, and it calculates a line at 26.8° where the data show almost nothing. Two weak lines, at 11.8° and 18.0°, belong to neither phase. $GdPt_2Ge$ formed in the structure that Alexandria predicts as its lowest-energy one, not in the *Cmcm* structure from OQMD that it was selected from.

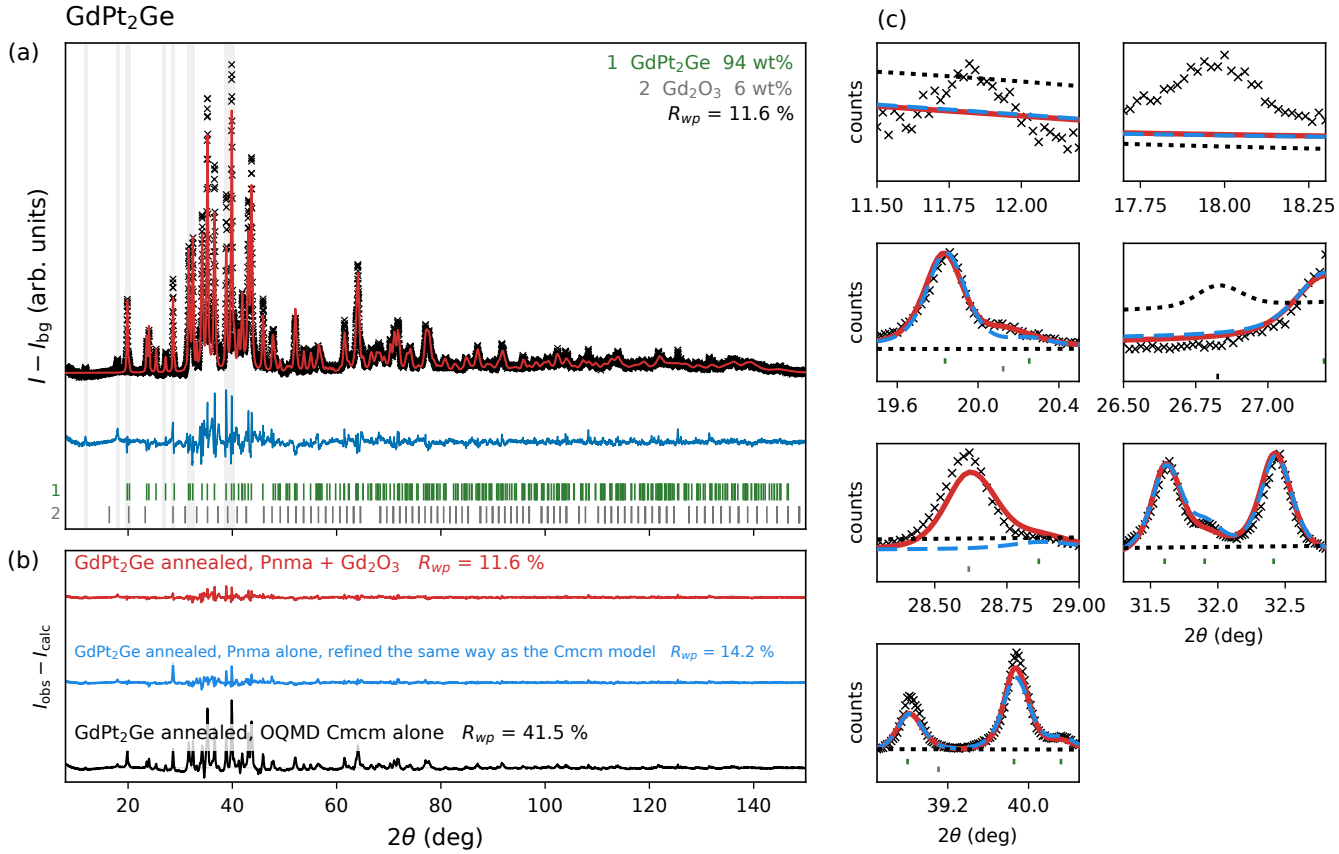


FIG. S4: Annealed $GdPt_2Ge$. Alternative models in (b): the *Pnma* phase alone, refined with exactly the same adjustable parameters as the *Cmcm* model, and the OQMD *Cmcm* structure alone. The top two windows in (c) hold the lines at 11.8° and 18.0° that no model explains. In the other windows, *Cmcm* (black dotted) calculates nothing at strong *Pnma* lines and a line near 26.8° that the data lack, and only the fit with the oxide reproduces the Gd_2O_3 (222) line.

4. $TiPt_5Si$

After annealing, 91 wt% of the $TiPt_5Si$ sample has the predicted $P4/mmm$ structure, and the remaining 9 wt% is an fcc phase (Fig. S5). A model with Ti, Si and Pt mixed on all sites gives the (110) line at 32.2° only a tenth of its observed intensity, so Pt is ordered on its own sites. This test cannot tell whether Ti and Si are also ordered. In the as-cast button, $TiPt_5Si$ was already the main phase (Sec. S4). The prediction is borne out: $TiPt_5Si$ formed in the predicted structure, with Pt ordered as predicted.

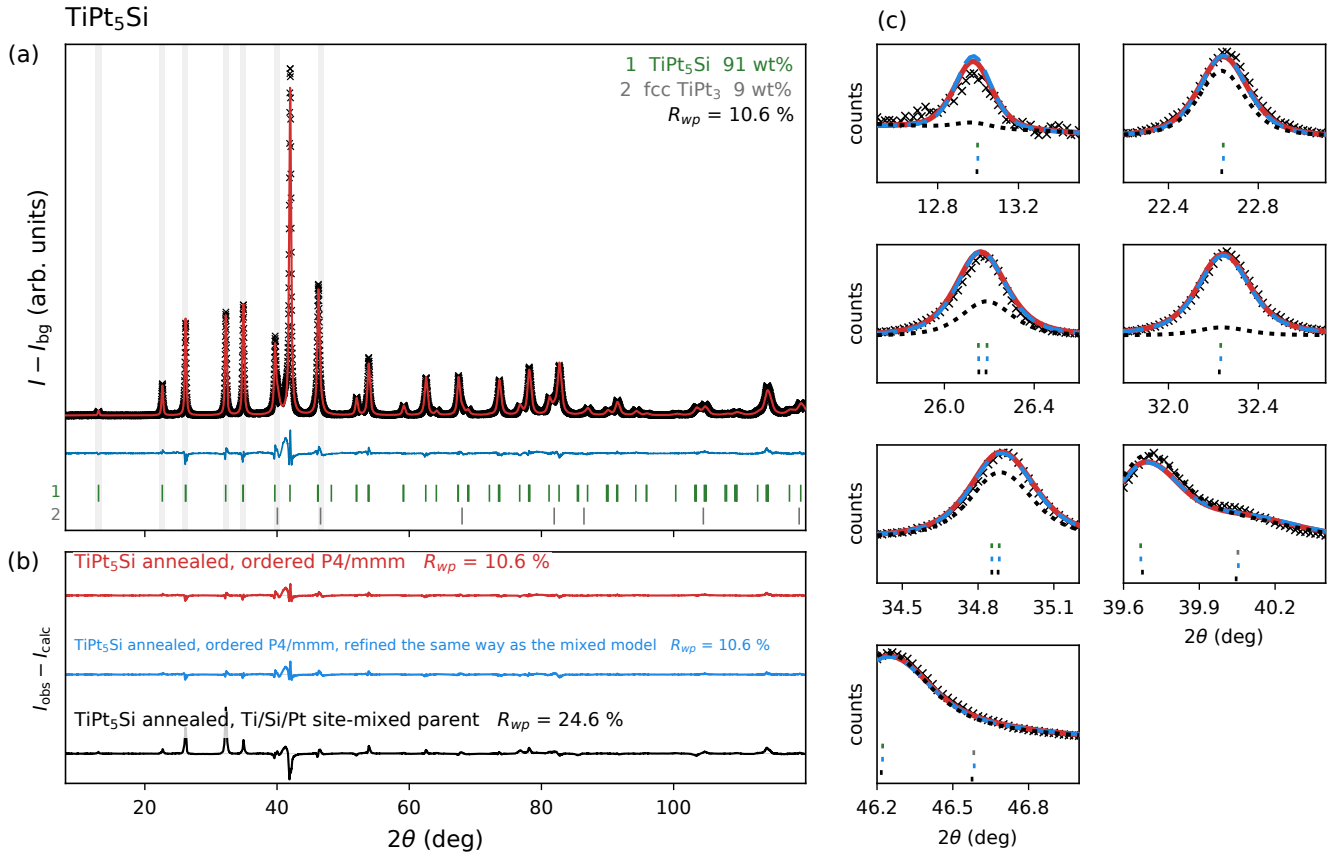


FIG. S5: Annealed $TiPt_5Si$, panels as in Fig. S2. The alternative model mixes Ti, Si and Pt on every site of the $P4/mmm$ cell. The alternative model loses the (001) and (110) superstructure lines in (c), and the windows near 40 and 46.5° show lines of the fcc phase (gray ticks) next to lines of $TiPt_5Si$ (green ticks).

5. *YSnCu*

Of the YSnCu sample, 90 wt% has the LiGaGe-type structure reported by Sebastian et al. [14], and the rest is likely $Y_3Cu_4Sn_4$ (Fig. S6). This structure is the lowest-energy structure in all three databases.

In this structure Cu and Sn occupy separate sites. A model with Cu and Sn mixed on both sites fits clearly worse (R_{wp} = 8.7% against 4.9%) and gives the (101) line at 25.9° less than half its observed intensity, so Cu and Sn are ordered. YSnCu formed in the predicted structure, but this compound had been reported before.

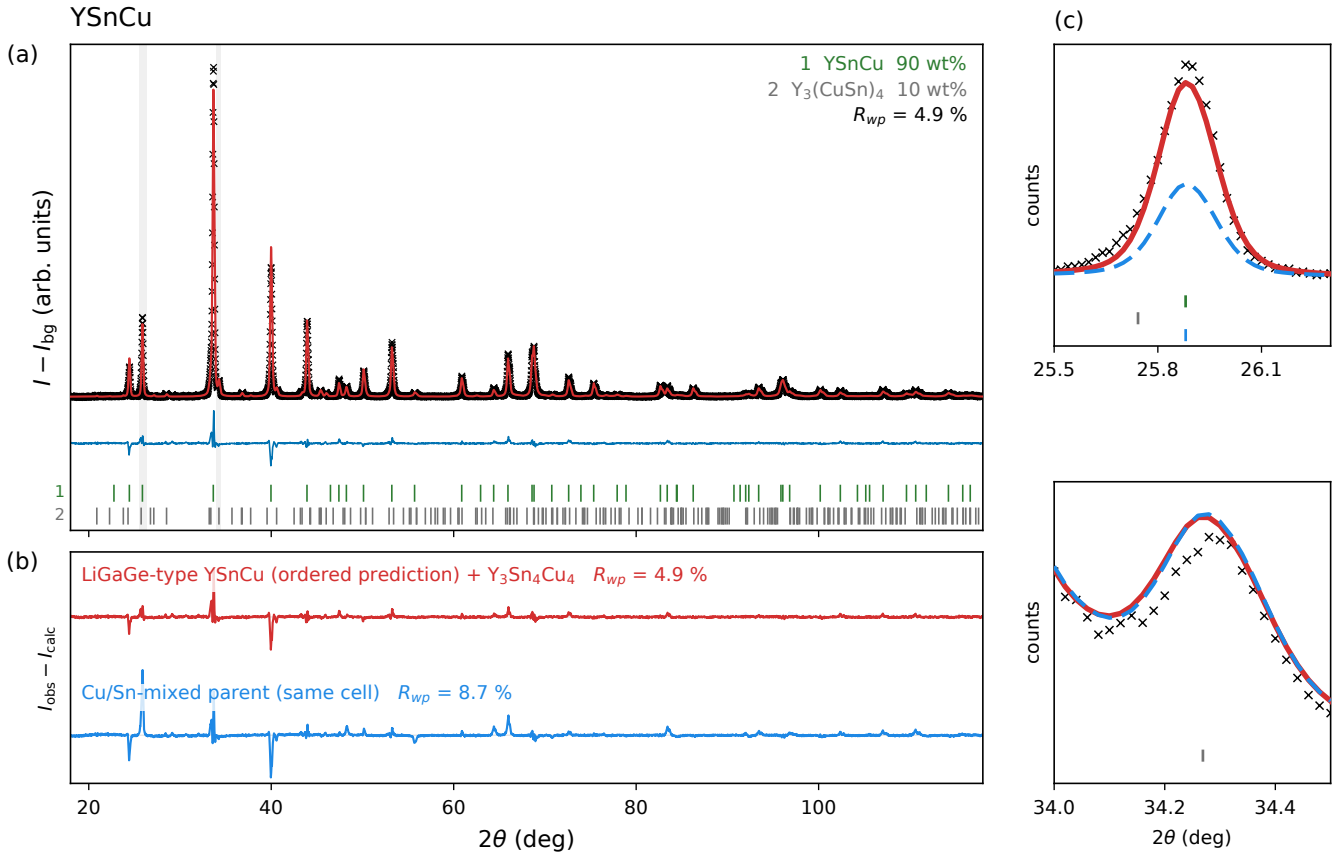


FIG. S6: As-cast YSnCu. The alternative model in (b) keeps the LiGaGe-type cell and Y site but puts Cu and Sn at random on the two remaining sites. Its (101) line, in the upper window of (c), is too weak by more than a factor of two. The lower window shows the strongest isolated line of $Y_3Cu_4Sn_4$.

6. $ScCrC_2$

About 95 wt% of the as-cast $ScCrC_2$ sample has the $P6_3/mmc$ structure of Pöttgen et al. [10], and the rest is likely graphite (Fig. S7). This structure is the lowest-energy structure in all three databases. Sc and Cr differ by only three in atomic number, but their order still shows in the (002) line at 19.4°. A model with Sc and Cr mixed on their sites gives this line less than half its observed intensity. The first button did not form the carbide, and the pattern shown is from a remake. A weak line at 35.9° is not explained by any phase we tried. $ScCrC_2$ formed in the predicted structure, but this compound had been reported before.

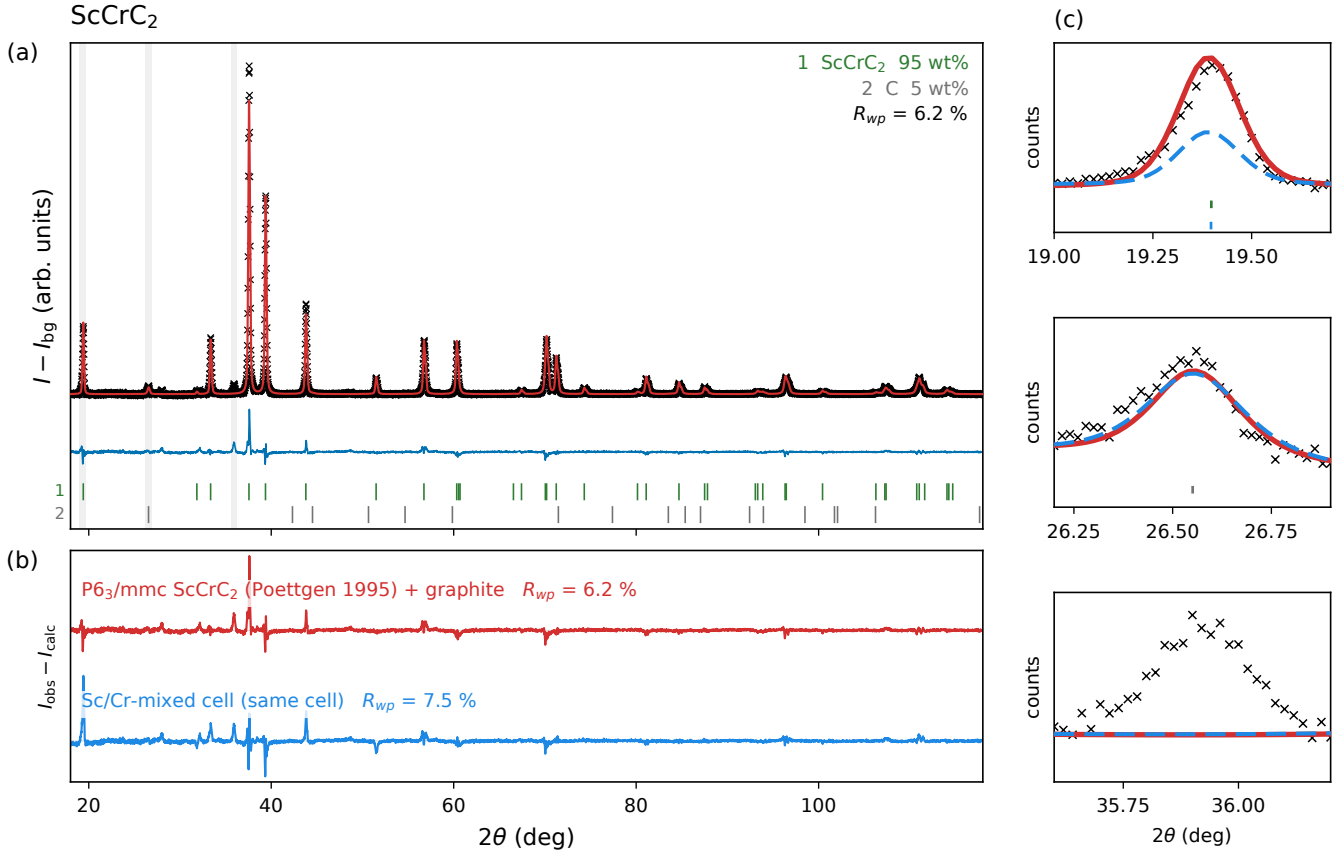


FIG. S7: As-cast $ScCrC_2$. The alternative model in (b) shares Sc and Cr equally between their two sites. The windows in (c) show the (002) line of $ScCrC_2$, which only the ordered model reproduces, the (002) line of graphite, and the unexplained line at 35.9°.

S7.2. Samples that formed an above-hull polymorph

1. *YCoSi*

After annealing, 93 wt% of the YCoSi sample has the *Pnma* structure, and the rest is likely YCo_2Si_2 (Fig. S8). Rogl [15] lists this structure for YCoSi. OQMD places the *Pnma* structure 32 meV/atom above the predicted $P4/nmm$ structure. The $P4/nmm$ structure alone fails at every strong line, with $R_{\rm wp} = 55\%$ against 10% for the fit with the *Pnma* structure and YCo_2Si_2. As cast, 75 wt% of the sample had the *Pnma* structure. The $P4/nmm$ structure did not fit the as-cast pattern either. YCoSi formed in a known structure that is higher in energy than the predicted one.

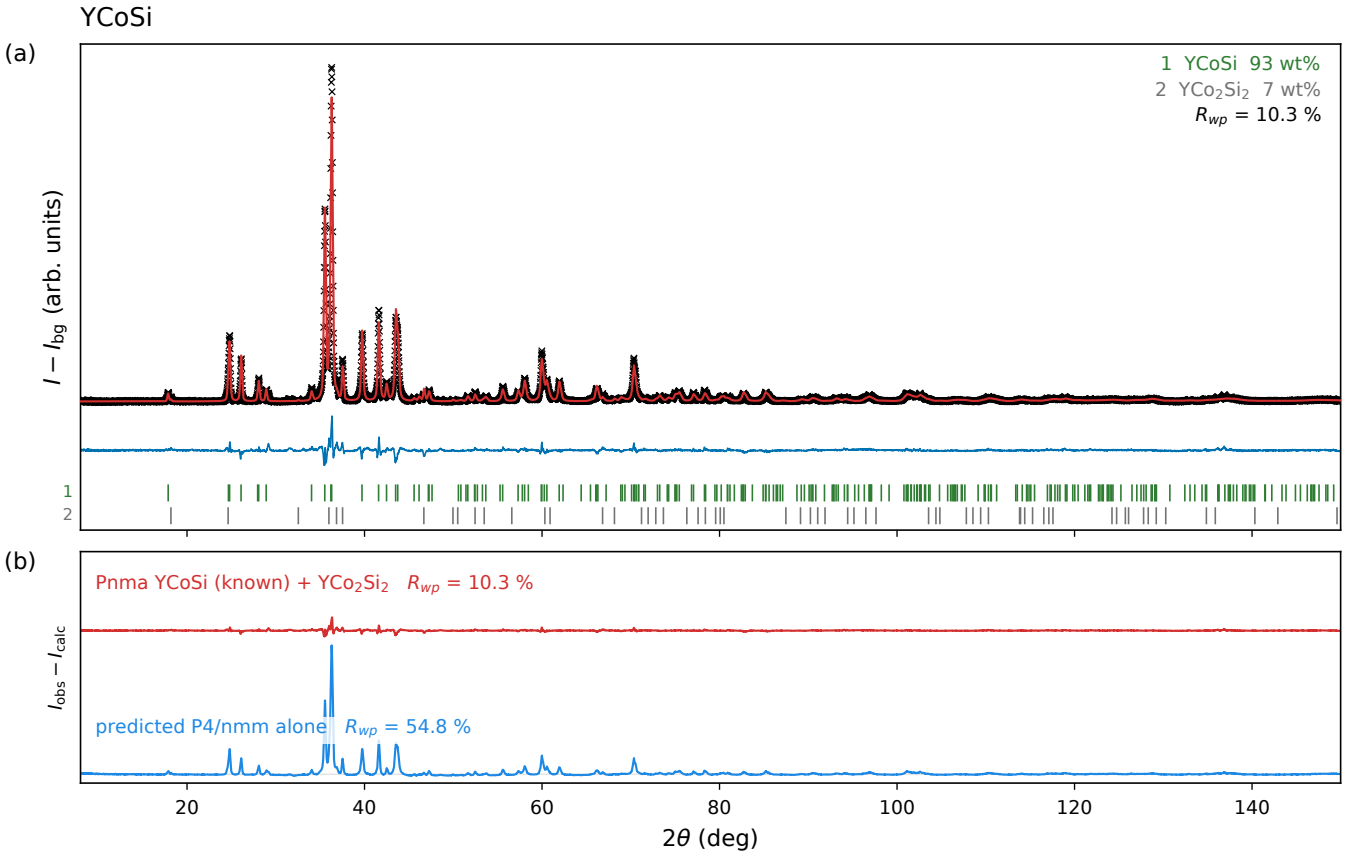


FIG. S8: Annealed YCoSi. In (b), red is the fit with the *Pnma* structure and YCo_2Si_2, and blue the fit with the predicted $P4/nmm$ structure alone. The $P4/nmm$ structure fails at every strong line. Thin-layer intensity correction refined (Sec. S3).

2. $YCoSi_3$

After annealing, the $YCoSi_3$ sample is a single phase with the $SmNiGe_3$-type $Cmmm$ structure (Fig. S9). OQMD places this structure 8 meV/atom above the predicted $I4mm$ structure, which it lists as the lowest-energy one. Bodak et al. [16] reported the composition but no structure. Two strong lines of the $I4mm$ structure, at 35.9° and 36.7°, would fall where the data are flat. A fit of the $I4mm$ structure alone diverged, so the figure shows the positions of its lines instead. $YCoSi_3$ formed in a structure that OQMD places slightly above the predicted one, not in the predicted structure.

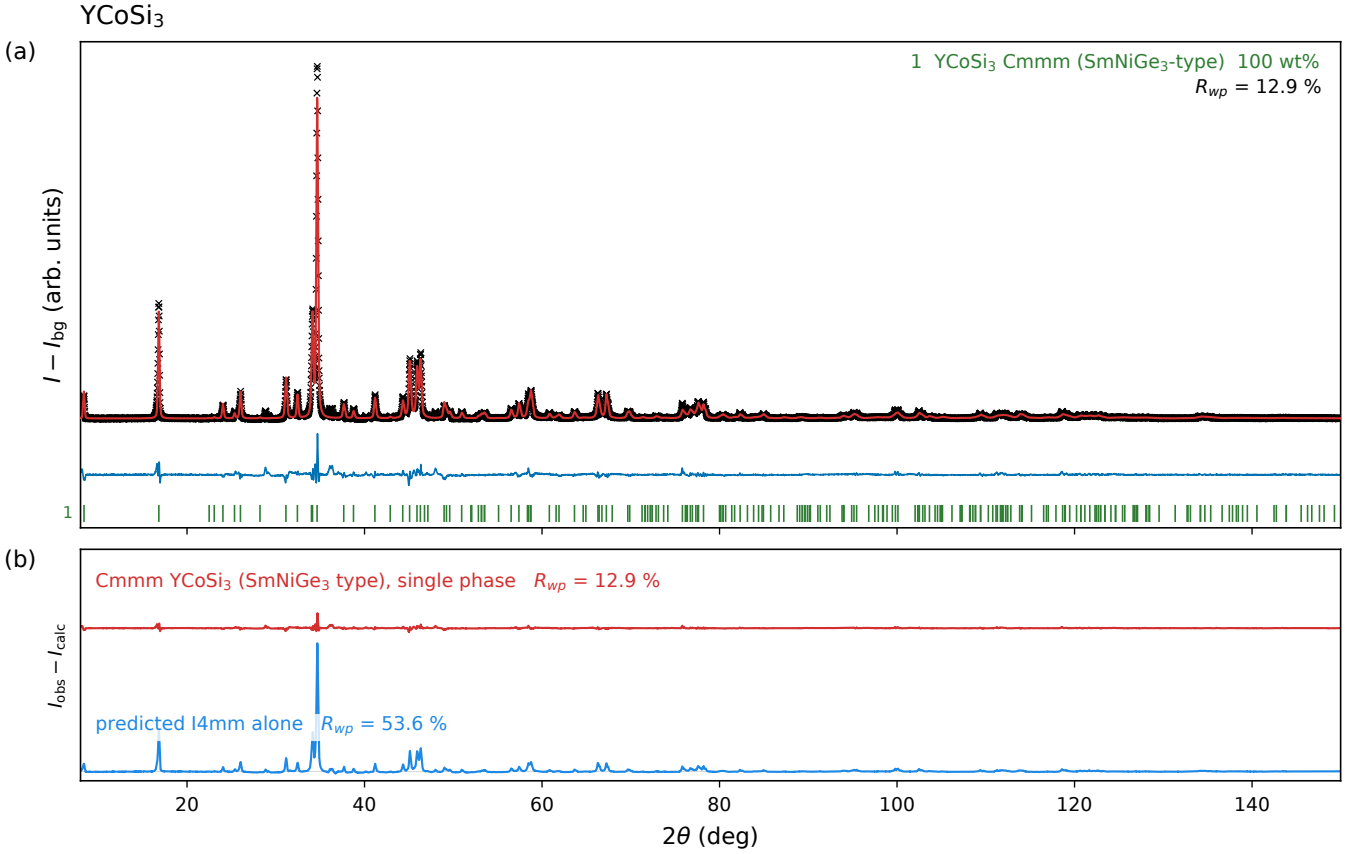


FIG. S9: Annealed $YCoSi_3$. In (b), red is the fit with the $Cmmm$ structure alone and blue the fit with the predicted $I4mm$ structure alone. The $I4mm$ structure fails at every strong line, so its difference curve shows the whole measured pattern. Thin-layer intensity correction refined (Sec. S3).

S7.3. Samples that formed a disordered parent of the on-hull structure

1. Al_8FeCo_7

Al_8FeCo_7 formed a B2 phase, with Al on one site and Fe and Co mixed on the other. The predicted structure is a $2 \times 2 \times 2$ B2 supercell with Fe and Co on separate sites. X-rays scatter almost equally from Fe and Co, so the lines produced by Fe/Co order are calculated at about 0.01% of the strongest line, far below the noise. The ordered model and the model with Fe and Co mixed both give $R_{\rm wp} = 5.9\%$, so our data cannot tell whether Fe and Co are ordered. Al_8FeCo_7 formed the predicted lattice with Al ordered, but our data cannot tell whether Fe and Co are also ordered as predicted, so both readings remain possible.

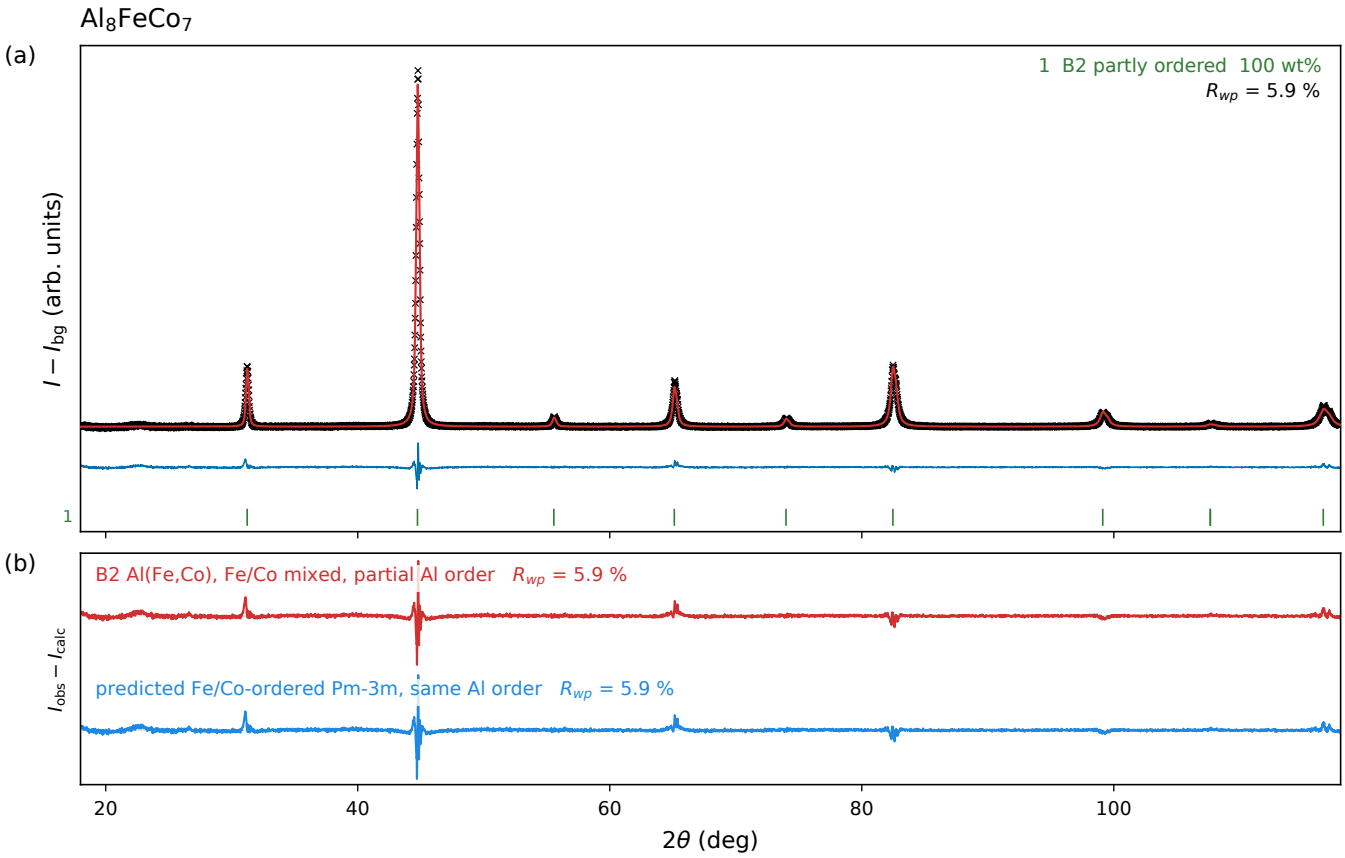


FIG. S10: As-cast Al_8FeCo_7. (b) The B2 model with Fe and Co mixed (red) and the predicted Fe/Co-ordered $Pm\bar{3}m$ model with the same Al order (blue) leave the same difference curve. The Fe/Co superlattice lines are too weak to show, so the figure has no panel (c).

2. Hf_2MoIr

After annealing, 88 wt% of the Hf_2MoIr sample is a bcc phase with partial Hf order, and 12 wt% is likely Ti_2Ni-type Hf_2Ir. The predicted $L2_1$ structure also orders Mo and Ir, which produces a (111) line at 23.6° calculated at 4% of the (110) line. The data show no line there (Fig. S11c). With the same Hf_2Ir phase, the $L2_1$ model gives $R_{wp} = 16.9\%$, against 7.3% for the fit of record. The B2 (100) line at 27.3° has less than half the intensity calculated for full Hf order, so the Hf order is partial. Hf_2MoIr formed the lattice of the predicted structure, but without the predicted Mo/Ir order and with only partial Hf order.

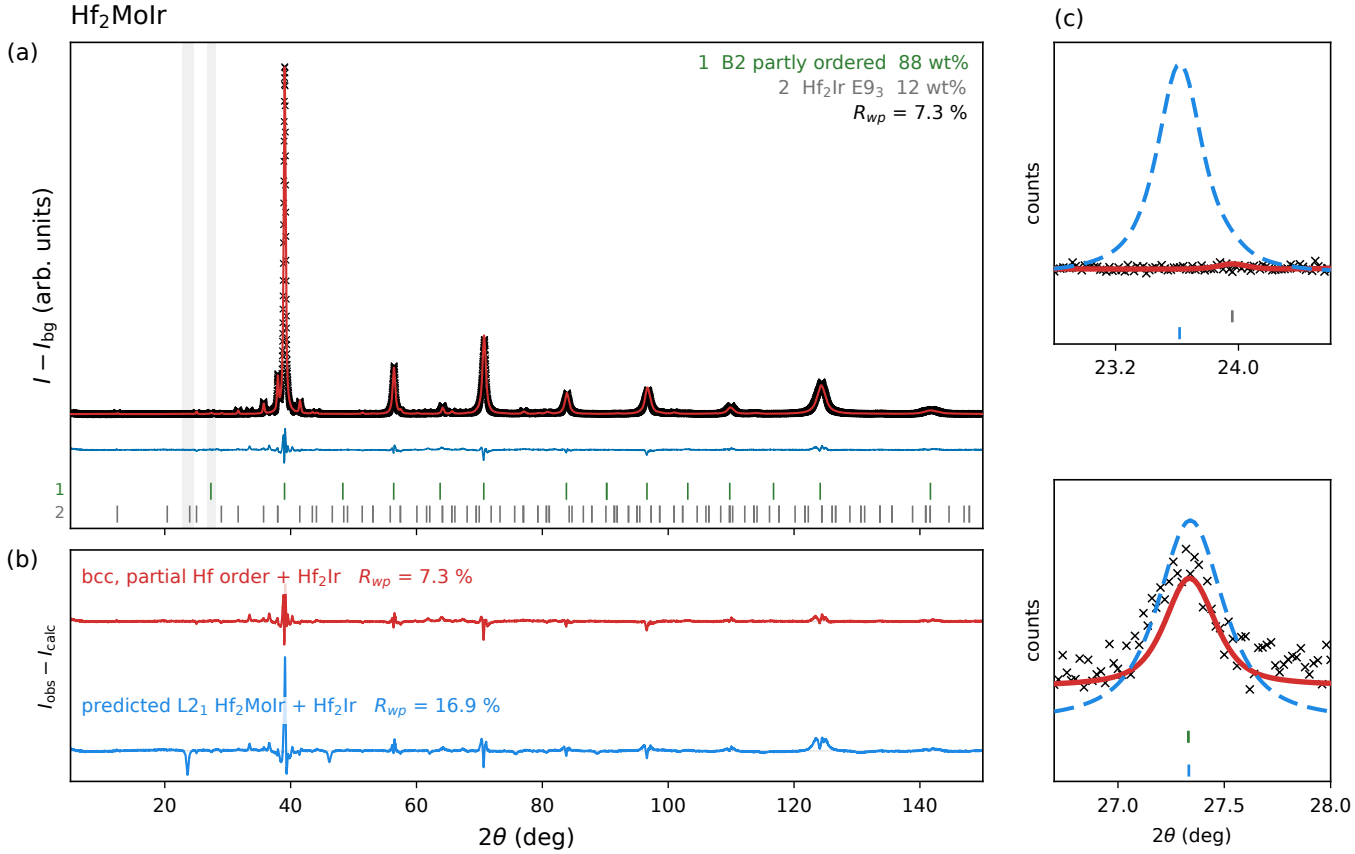


FIG. S11: Annealed Hf_2MoIr. (b) The bcc model with partial Hf order (red) and the predicted $L2_1$ structure (blue), each with Hf_2Ir. (c) The $L2_1$ (111) line at 23.6°, calculated by the prediction and absent from the data, and the B2 (100) line at 27.3°, which full Hf order over-calculates.

3. Al_4FeCo_3

Al_4FeCo_3 formed a B2 phase, with Al on one site and Fe and Co mixed on the other. The predicted structure is a body-centered $Im\bar{3}m$ supercell with Fe and Co on separate sites. As in Al_8FeCo_7, the lines produced by Fe/Co order are far too weak to detect with X-rays, and the ordered and Fe/Co-mixed models both give $R_{\rm wp} = 5.8\%$. Neutron diffraction could settle the Fe/Co order in both compounds, since the neutron scattering lengths of Fe and Co differ by almost a factor of four. Al_4FeCo_3 formed the predicted lattice with Al ordered, but our data cannot tell whether Fe and Co are also ordered as predicted, so both readings remain possible.

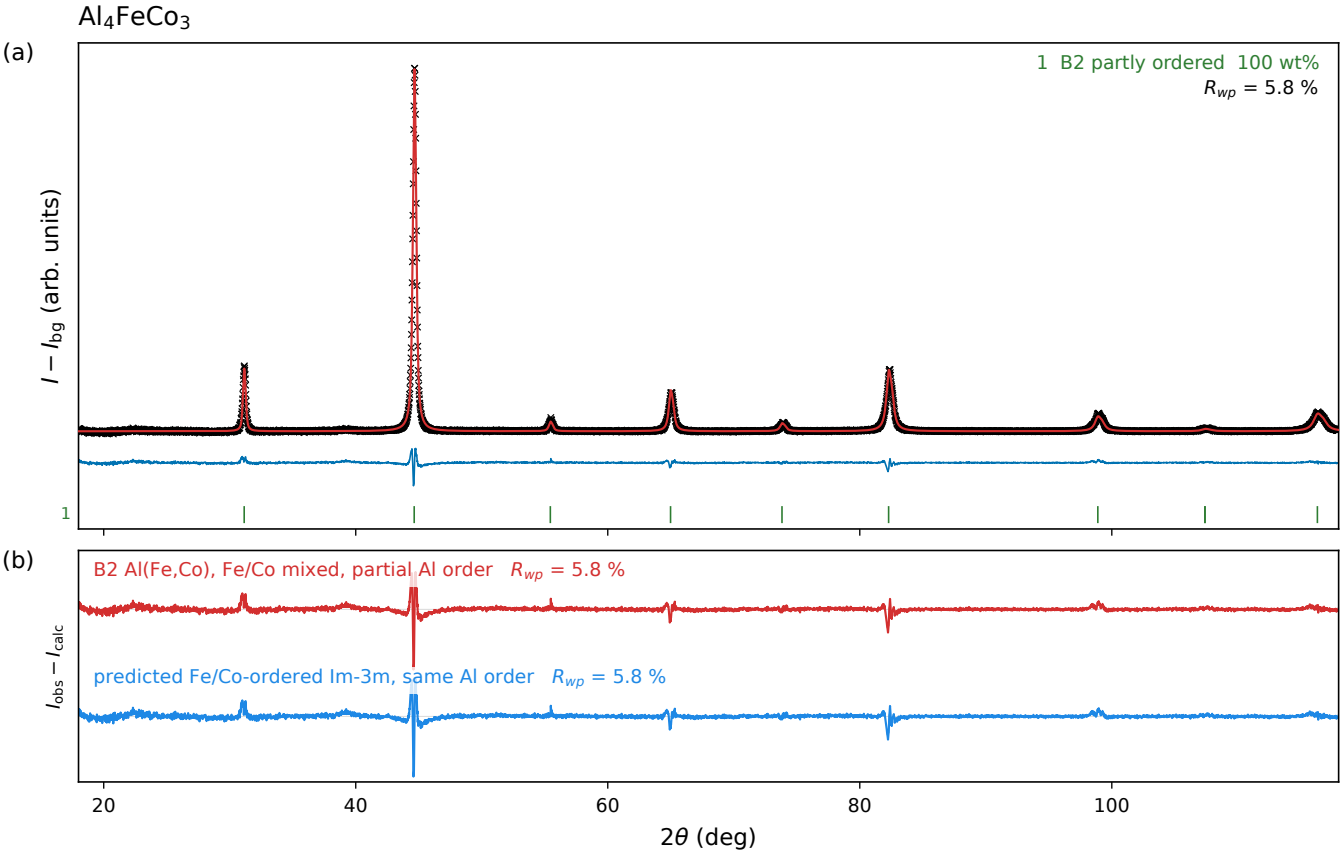


FIG. S12: As-cast Al_4FeCo_3. (b) The B2 model with Fe and Co mixed (red) and the predicted Fe/Co-ordered $Im\bar{3}m$ model with the same Al order (blue), which leave the same difference curve. As for Al_8FeCo_7, the Fe/Co superlattice lines are too weak to show.

4. Sc_2PdCu

The predicted structures do not match the pattern of Sc_2PdCu. The lowest-energy structures, $L2_1$ in OQMD and MP and $P4/mmm$ in Alexandria, both order Cu and Pd, which adds superlattice lines to the B2 pattern. The $L2_1$ (111) line at 23.6° is calculated at 8% of the strongest line, and the $P4/mmm$ (101) line at 30.7° at 4%, but the data are flat at both positions (Fig. S13c). The best match we found is a B2 phase, with Sc on one site and Cu and Pd mixed on the other ($R_{\rm wp} = 8.1\%$). The pattern is from a cut face of the as-cast button. The B2 (100) line at 27.2° is weaker than full Sc order calculates, so the Sc order is probably incomplete. So the predicted Cu/Pd order did not form.

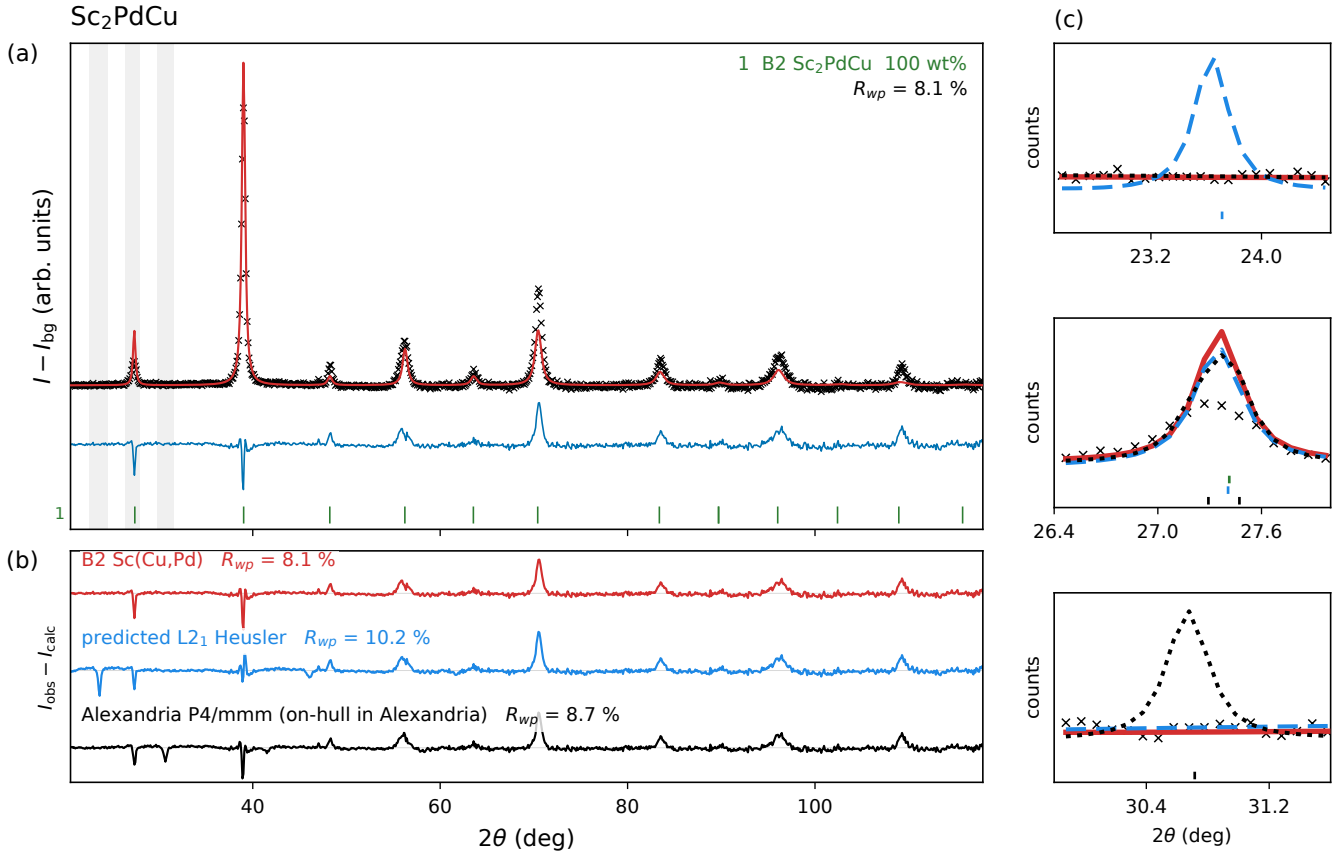


FIG. S13: As-cast Sc_2PdCu, bulk cut face. (b) B2 with full Sc order (red), the predicted $L2_1$ Heusler structure (blue) and the Alexandria $P4/mmm$ structure (black). (c) The $L2_1$ (111) line at 23.6°, the B2 (100) line at 27.2° and the $P4/mmm$ (101) line at 30.7°. The ordered structures each calculate a line that the data lack.

5. Sc_3Pd_2Cu

The predicted structure does not match the pattern of Sc_3Pd_2Cu. It has Cu and Pd in alternate layers, which produces a (001) line at 8.95° calculated at 6% of the strongest line. We started this refinement at 8° to include that line, and the line is absent, even after a week at 1000 °C (Fig. S14c). The best match we found is a B2 phase, with Sc partly ordered and Cu and Pd mixed ($R_{\rm wp} = 4.3\%$). So the predicted Cu/Pd order did not form, although the B2 phase has the same underlying lattice as the predicted structure.

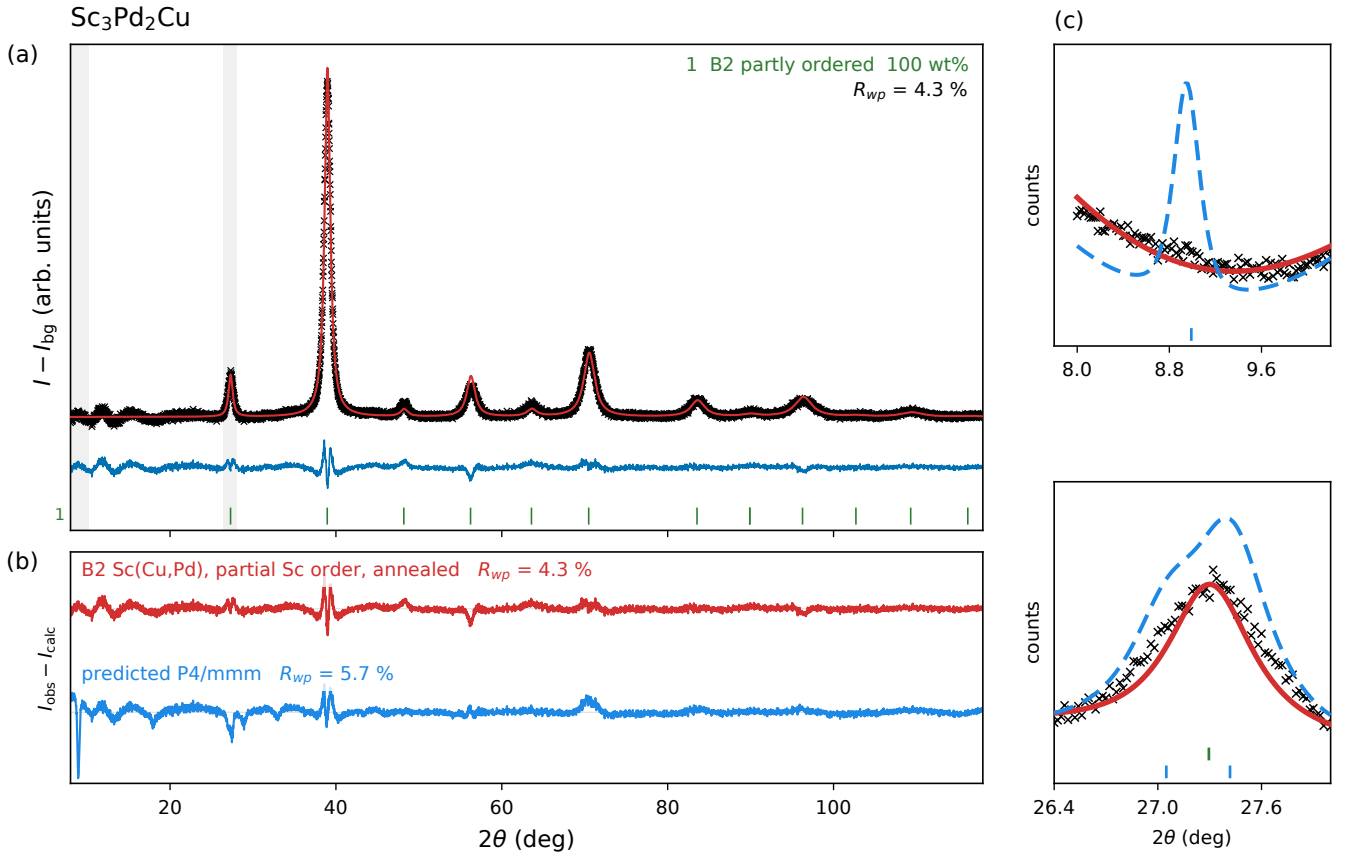


FIG. S14: Annealed Sc_3Pd_2Cu. (b) B2 with partial Sc order (red) and the predicted $P4/mmm$ structure (blue). (c) The $P4/mmm$ (001) line at 8.95° and the B2 (100) line at 27.2°. The blue curve includes that fit's own background, which bends below the data around the missing line.

6. $ScPd_5Au_2$

The fcc lattice matches every strong line of $ScPd_5Au_2$, but the intensities of several lines are poorly fitted (R_{wp} = 15.3%), because the pattern is from the uncut face of the as-cast button rather than a powder (Fig. S15). The predicted *Pbam* structure is an ordered version of the same fcc lattice. Within the measured range it differs from the disordered fcc model mainly by weak superlattice lines, the strongest near 32°, calculated at about 5% of the strongest line. The data show at most a weak bump near 31.6°, slightly below the calculated position, so the conclusion rests on this one weak feature. The scan starts at 30.5°, so the stronger superlattice lines at lower angle were not measured, and the button lost 16% of its mass during melting. We take the predicted order as not observed, but it is not excluded, and the conclusion for this sample is provisional.

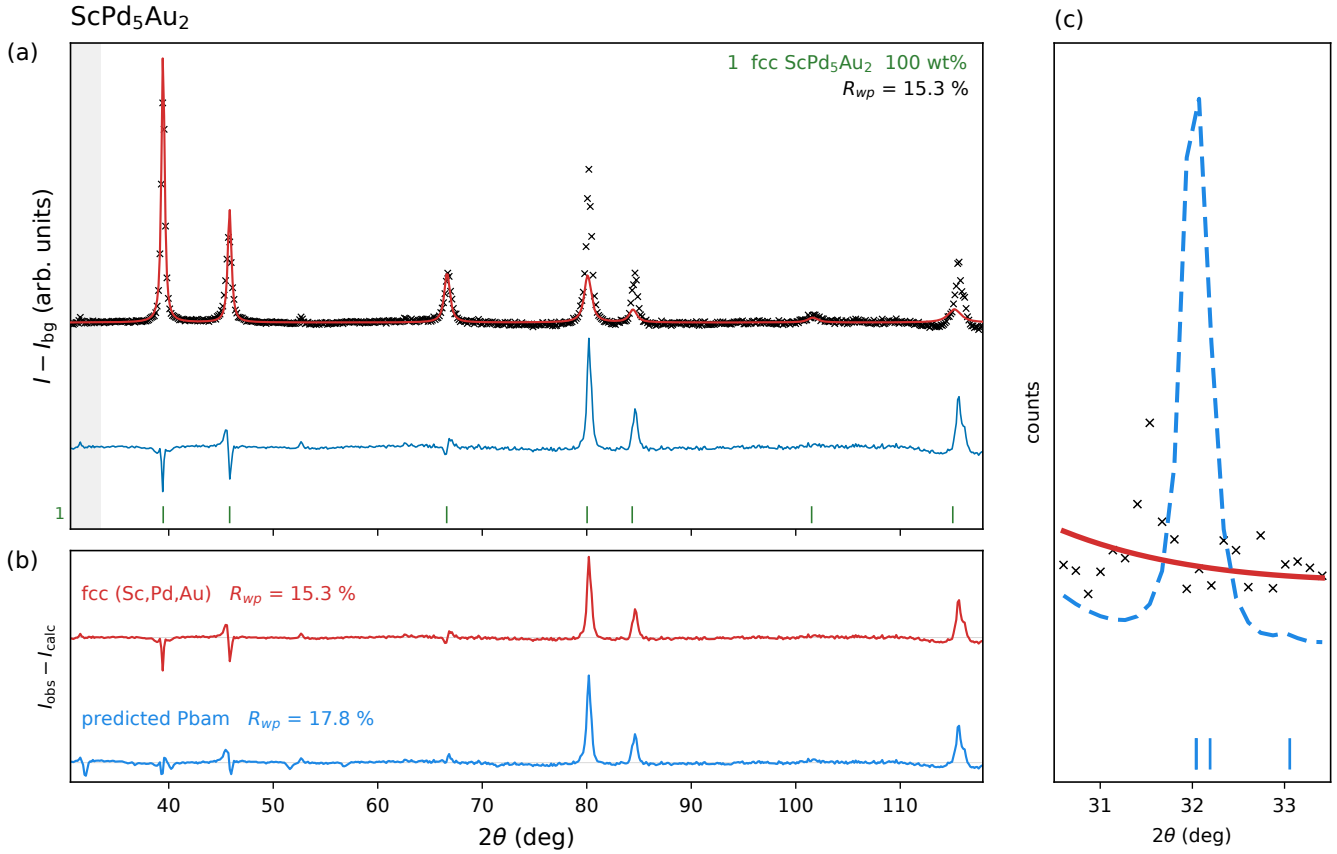


FIG. S15: As-cast $ScPd_5Au_2$, uncut button face. In (b) and (c), red is the disordered fcc model and blue dashed the predicted *Pbam* structure. (c) The region of the strongest *Pbam* superlattice line in the measured range, near 32°, where the data show at most a weak bump near 31.6°.

7. Al_2MnAu_6

The best fit to Al_2MnAu_6 uses a $C11_b$ cell in which Au is only partly ordered, so that a third of the site shared by Al and Mn also holds Au. The predicted $C2/m$ structure orders Mn and Al on that site. With the same partial Au order, the $C2/m$ model fits no worse ($R_{wp} = 5.9\%$ against 6.2%). The lines produced only by Mn/Al order, near 32.3°, are calculated at about 10 counts, well below the scatter of the data. The small improvement of the $C2/m$ model comes from the extra freedom of its lower, monoclinic symmetry in fitting the broad lines, not from these lines. Our data therefore cannot tell whether Mn and Al in Al_2MnAu_6 are ordered as predicted, so both readings remain possible.

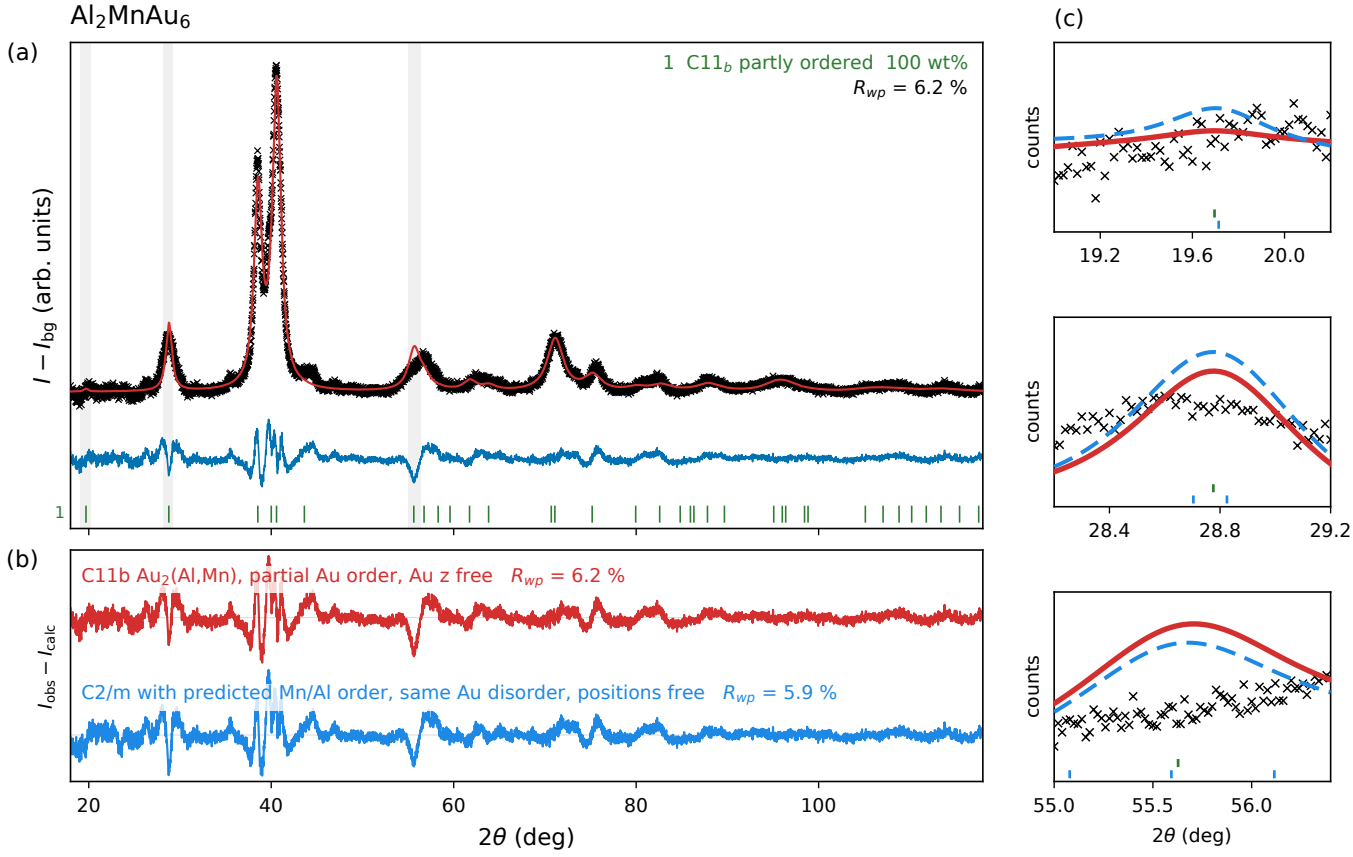


FIG. S16: As-cast Al_2MnAu_6. (b) $C11_b$ with Al and Mn mixed (red) and $C2/m$ with the predicted Mn/Al order (blue), both with the same partial Au order. (c) The $C11_b$ (002) and (101) lines near 19.6° and 28.7° and the broad line near 55.6°. The two models misplace the lines at 28.7° and 55.6° by the same amount. Sample displacement held at zero (Sec. S3).

8. Sc_2Pd_5Au

The predicted structures do not match the pattern of Sc_2Pd_5Au. The *Pbam* structure, the lowest-energy structure in OQMD and Alexandria, orders Pd and Au, which produces weak lines near 15.7° and 17.6°. These lines are calculated at five to ten times the noise, but neither appears in the pattern (Fig. S17c). The *Amm*2 structure, the lowest-energy structure in MP, does not match either, since its strong lines at 17.9° and 26.4° fall where the data are flat. The best match we found is an $L1_2$ $Sc(Pd,Au)_3$ phase, with Sc on the cube corners and Pd and Au mixed on the face centers ($R_{wp} = 7.2\%$).

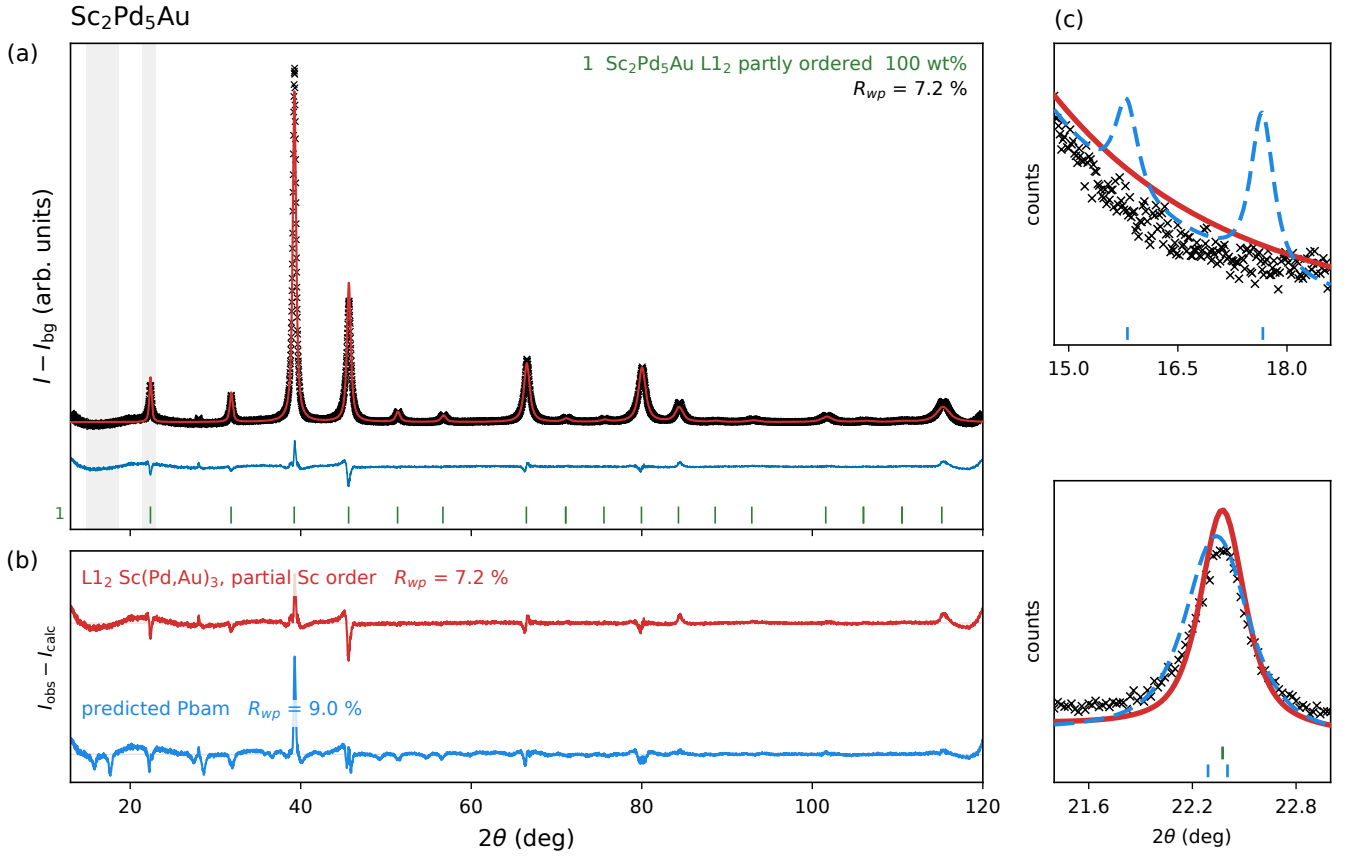


FIG. S17: As-cast Sc_2Pd_5Au. (b) $L1_2$ with Sc order and Pd and Au mixed (red) and the predicted *Pbam* structure (blue). (c) The *Pbam* lines near 15.7° and 17.6° and the $L1_2$ (100) line at 22.3°. Below 17.5° the fitted background runs above the data in both fits, so the absence of the *Pbam* lines is judged from the shape of the data, not from the difference curve.

9. $PtNiGe_2$

The predicted structure does not match the pattern of $PtNiGe_2$, but the fit is poor ($R_{wp} = 21\%$), so this conclusion is provisional. The predicted $P2_1/m$ structure orders Ni and Pt, and its strongest order line, (100) at 15.9°, is calculated at more than half the intensity of its strongest line. We started both refinements at 14° to include this line, and the data there show a smooth tail with no peak. A weaker order line at 39.0° is also absent. The best match we found is MnP-type (Ni,Pt)Ge with Ni and Pt mixed, which at equal amounts of Ni and Pt has the nominal composition. A strong line at 32.2°, about 70% of the strongest observed line, is not explained by any model we tried.

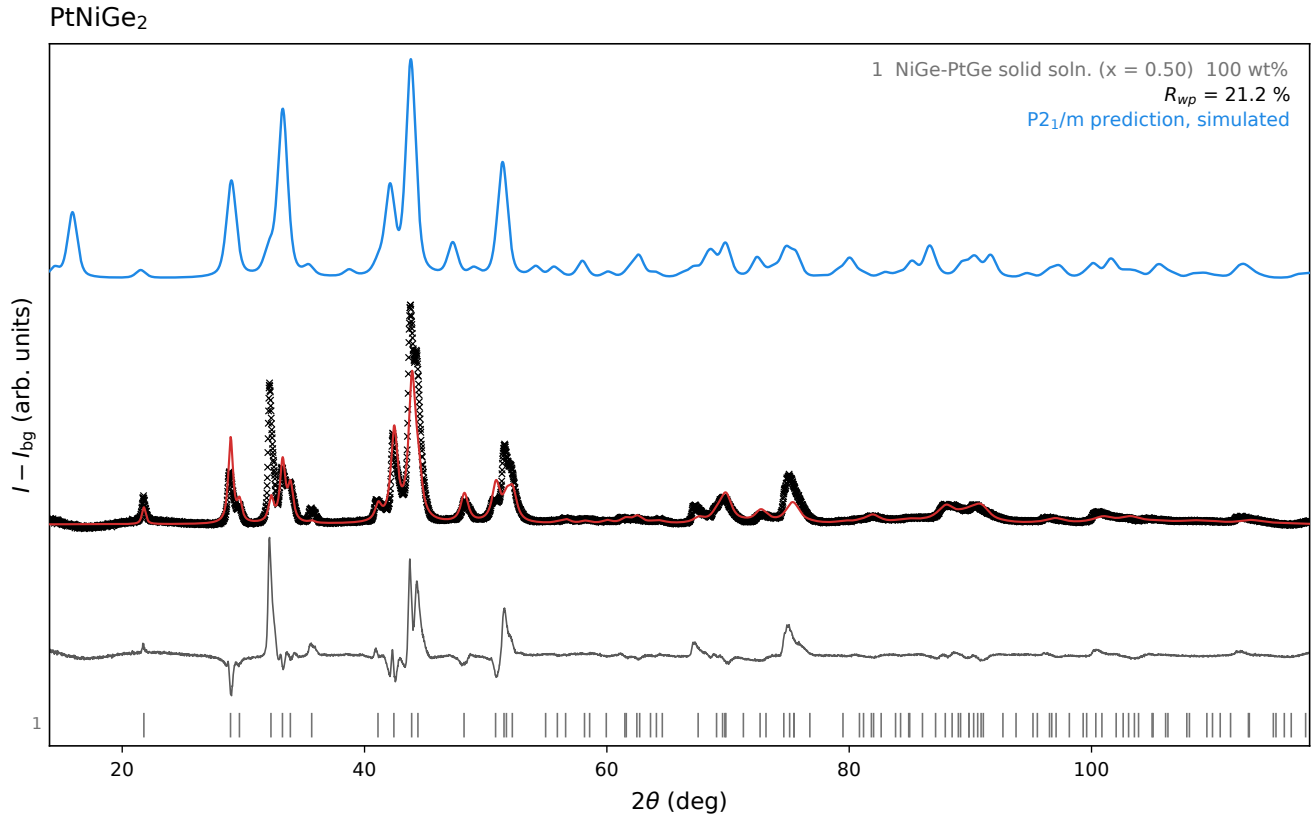


FIG. S18: $PtNiGe_2$. Measured pattern with the background subtracted (black crosses), the best fit (red) and its residual (gray), with the Bragg positions of its phases below. Above them, offset upward, the pattern of the predicted structure simulated from its computed crystal structure, with line widths taken from the measured pattern (blue), scaled to the strongest observed line.

10. Gd_3Sn_6Co

The predicted structure does not match the pattern of Gd_3Sn_6Co. The structure OQMD predicts is a *Cmcm* structure with ordered Co, which shares its strong lines with the known $CeNiSi_2$-type $GdCo_xSn_2$ and fits about as well overall. The line that decides between them is the *Cmcm* (152) line at 36.5°, which comes from the Co order. This line is calculated at about forty times the noise, and the data show no line there (Fig. S19c). The best match we found is $GdCo_xSn_2$ alone, in which Co partly fills one site, refined with the structure of Gil et al. [9] ($R_{\rm wp} = 13.4\%$, Fig. S19). So the predicted Co order did not form.

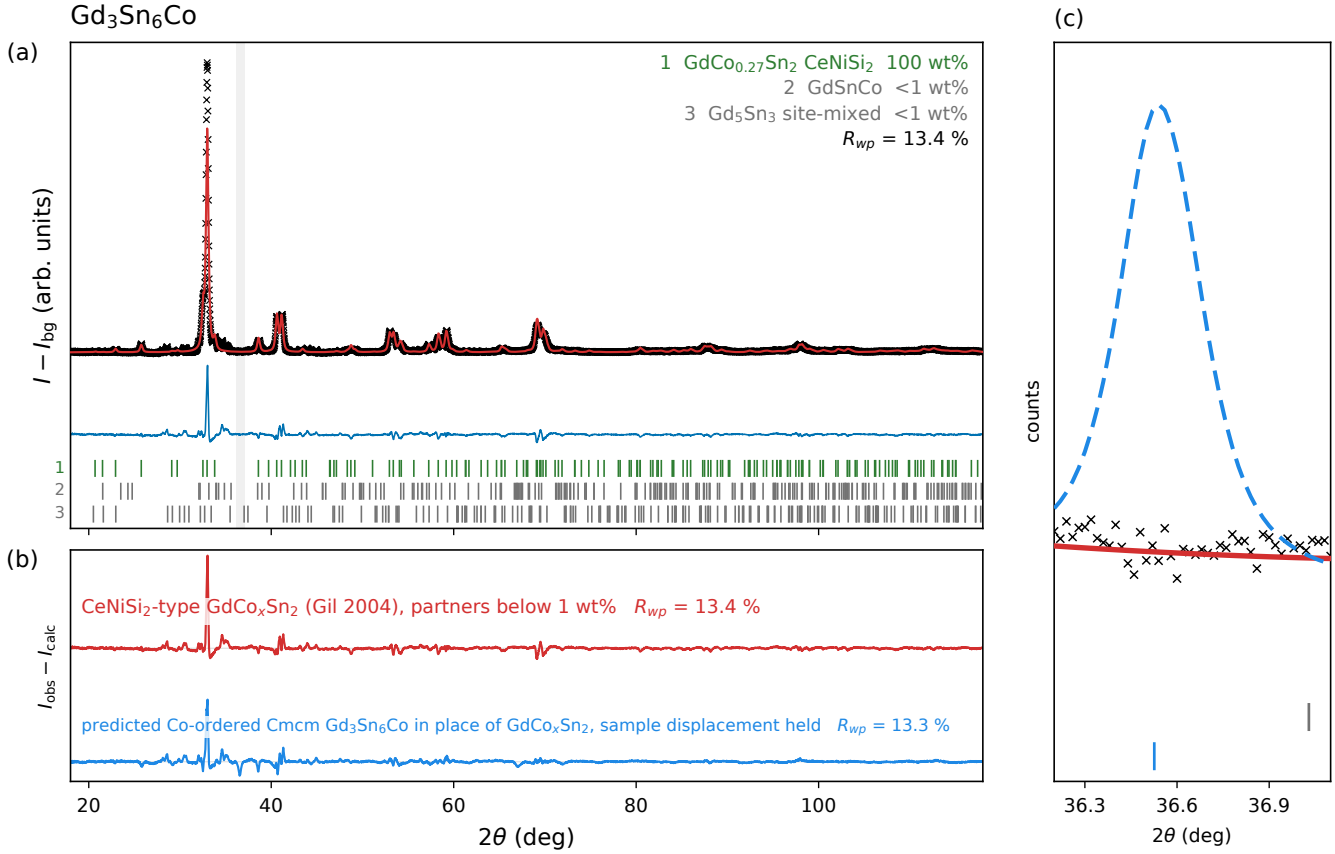


FIG. S19: As-cast Gd_3Sn_6Co. (b) $CeNiSi_2$-type $GdCo_xSn_2$ (red) and the predicted Co-ordered *Cmcm* structure in its place (blue). (c) The (152) line of the prediction at 36.5°, absent from the data.

S7.4. Samples that formed another solid solution

1. $ScPd_2Au$

The lines of $ScPd_2Au$ are broad, which hides much of the detail, and our best fit describes the pattern only approximately ($R_{wp} = 7.9\%$, Fig. S20). That fit uses two $Sc(Pd,Au)_3$ solid solutions that differ in their stacking, a hexagonal $D0_{24}$-type phase (about 58 wt%) and a cubic $L1_2$ phase with Pd and Au mixed on the face centers. Other combinations of disordered phases may describe the pattern as well. The predicted $P4/mmm$ structure clearly does not fit. With its DFT cell it gives $R_{wp} = 18.6\%$, because its tetragonal cell splits the $L1_2$ (200) line into two lines, at 44.4 and 46.5°, where the data show a single line at 45.5°. With the predicted Pd/Au order placed on the cubic cell, the superlattice line at 31.6° is calculated too strong ($R_{wp} = 8.5\%$), so full Pd/Au order is unlikely. A hexagonal line overlaps that position, so partial order cannot be excluded. The predicted structure did not form, and the best fits come from disordered solid solutions.

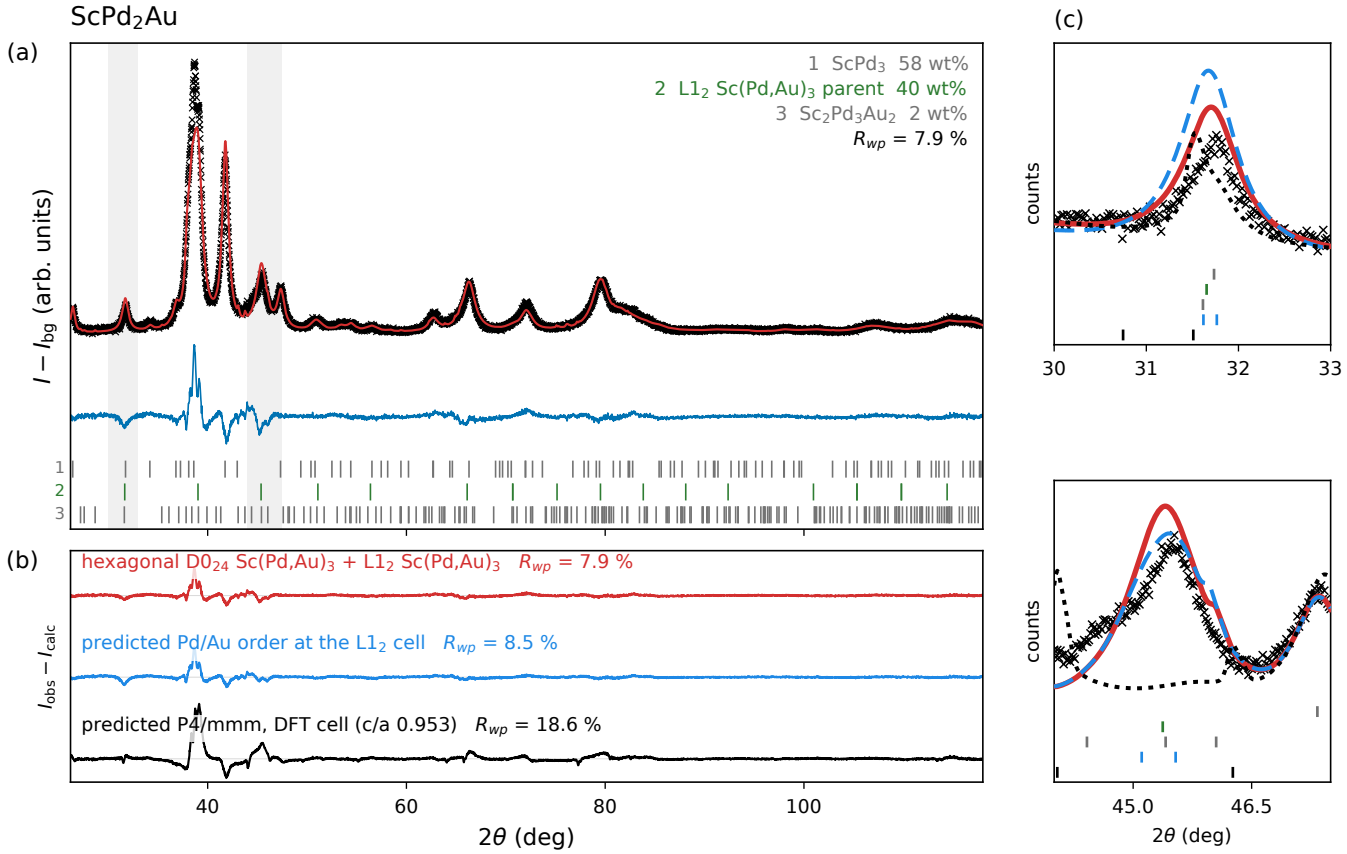


FIG. S20: $ScPd_2Au$. (c) The superlattice line near 31.6° and the (200) line near 45.5°. Sample displacement held at zero (Sec. S3).

S7.5. Samples that separated into several phases, none of them the on-hull structure

1. $ScPd_2Cu$

The predicted $D0_{22}$ structure does not match the pattern of $ScPd_2Cu$ well, and the fit we found is poor, so the description of what formed is provisional (Fig. S21). Refined with lattice parameters near its DFT values, the $D0_{22}$ model fits much worse than our best match ($R_{wp} = 12.8\%$ against 6.1%). Allowed to change its cell further, it becomes nearly cubic, so the data rule out the predicted cell shape but not the predicted Cu/Pd order. The best match we found has two phases, an fcc-derived $Sc(Cu,Pd)_3$ phase with Cu and Pd mixed (about 61 wt%) and a hexagonal $CuPd_3$-type phase. A model with Cu and Pd ordered on the same fcc-derived lattice fits about as well ($R_{wp} = 6.8\%$), because the lines are broad. A sharp line at 91.4° is not explained by any model we tried. The hexagonal model contains no Sc, so the composition of the fcc-derived phase is not known, and if it has the nominal composition it is a disordered version of the predicted structure.

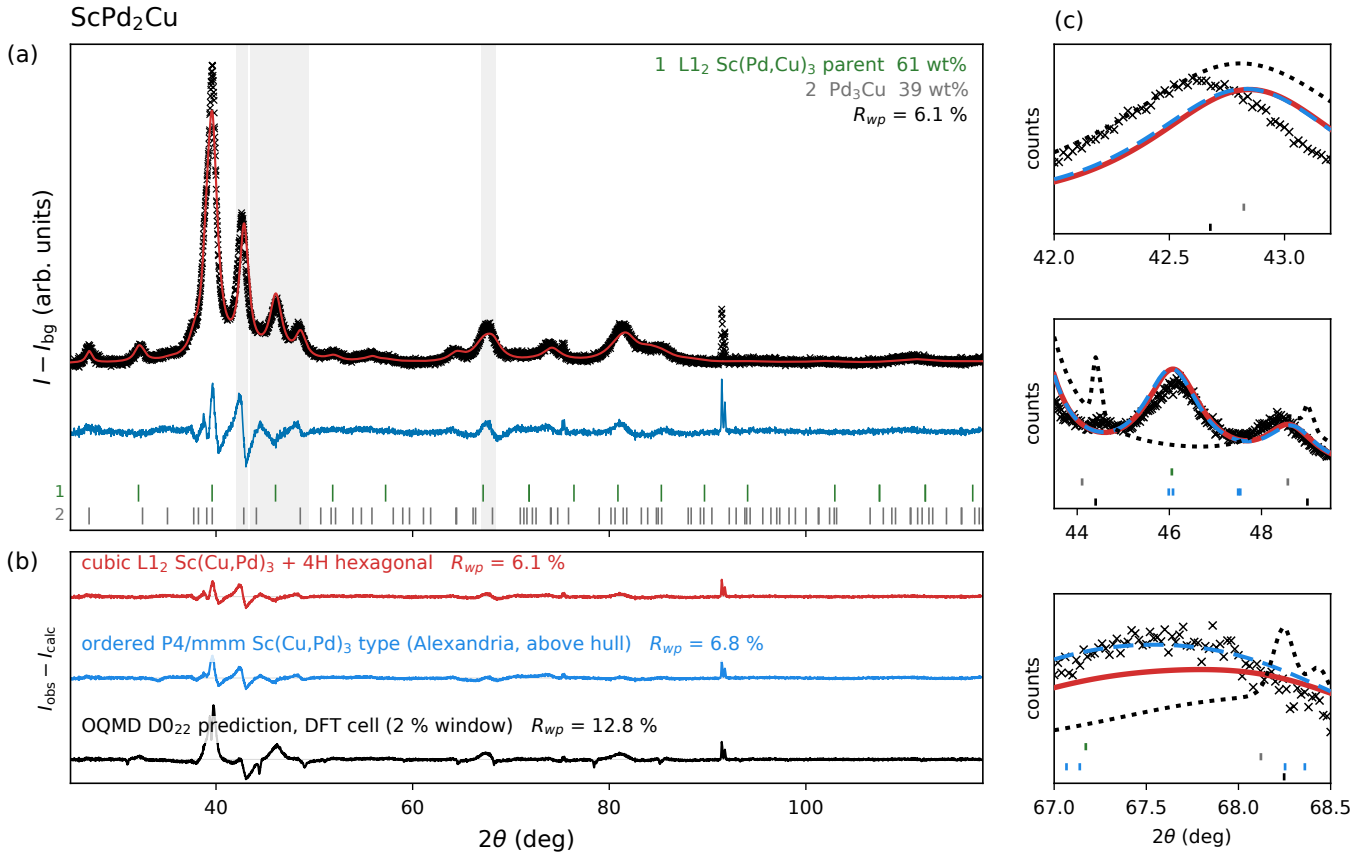


FIG. S21: $ScPd_2Cu$. (c) Lines near 42.6 and 46.2° and the (220) line. The lines are broad, and the disordered and ordered fcc-derived models differ little. Sample displacement held at zero (Sec. S3).

2. $La_5Al_3Co_2$

The best match we found to the strongest lines of $La_5Al_3Co_2$ is the known $La_4Co_3Al_3$, refined with the structure of Tursina et al. [12] (Fig. S22). Its presence is robust, because it accounts for the strongest lines. It cannot be the whole sample, because its composition differs from the nominal one, and some lines are not explained by $La_4Co_3Al_3$, but we have not identified the other phases. The predicted $Cmcm$ structure was not identified.

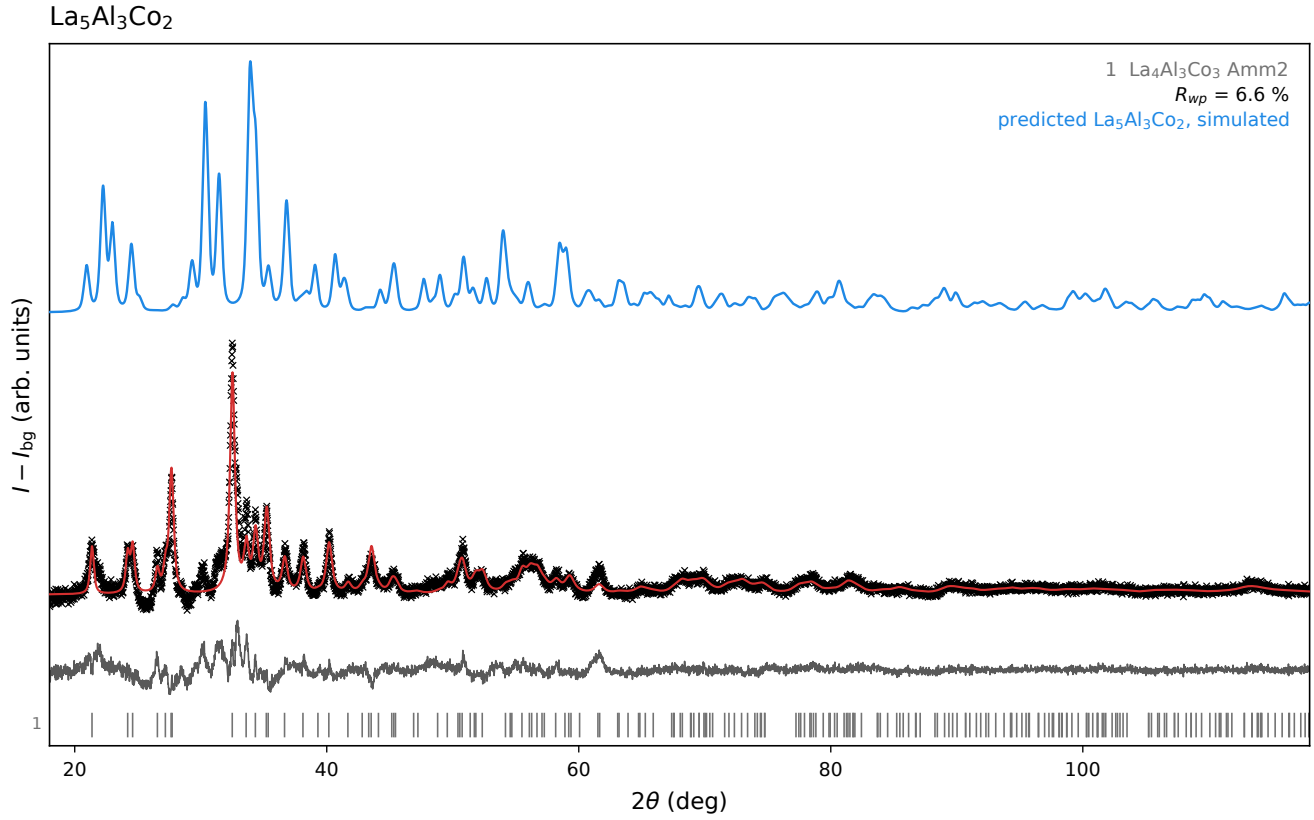


FIG. S22: $La_5Al_3Co_2$. Measured pattern with the background subtracted (black crosses), the best fit (red) and its residual (gray), with the Bragg positions of its phases below. Above them, offset upward, the pattern of the predicted structure simulated from its computed crystal structure, with line widths taken from the measured pattern (blue), scaled to the strongest observed line.

3. $TiPt_5Ge$

The predicted structure does not match the pattern of $TiPt_5Ge$. Refined alone, the predicted structure gives $R_{wp} = 22.7\%$, and its line at 31.66°, clear of the Pt and Pt_2Ge lines, is absent from the data (Fig. S23). A mixture of fcc Pt and Fe_2P-type Pt_2Ge matches the pattern well ($R_{wp} = 5.4\%$). Neither phase contains Ti, so a Ti-bearing phase, probably minor or poorly crystallized, is missing from the model.

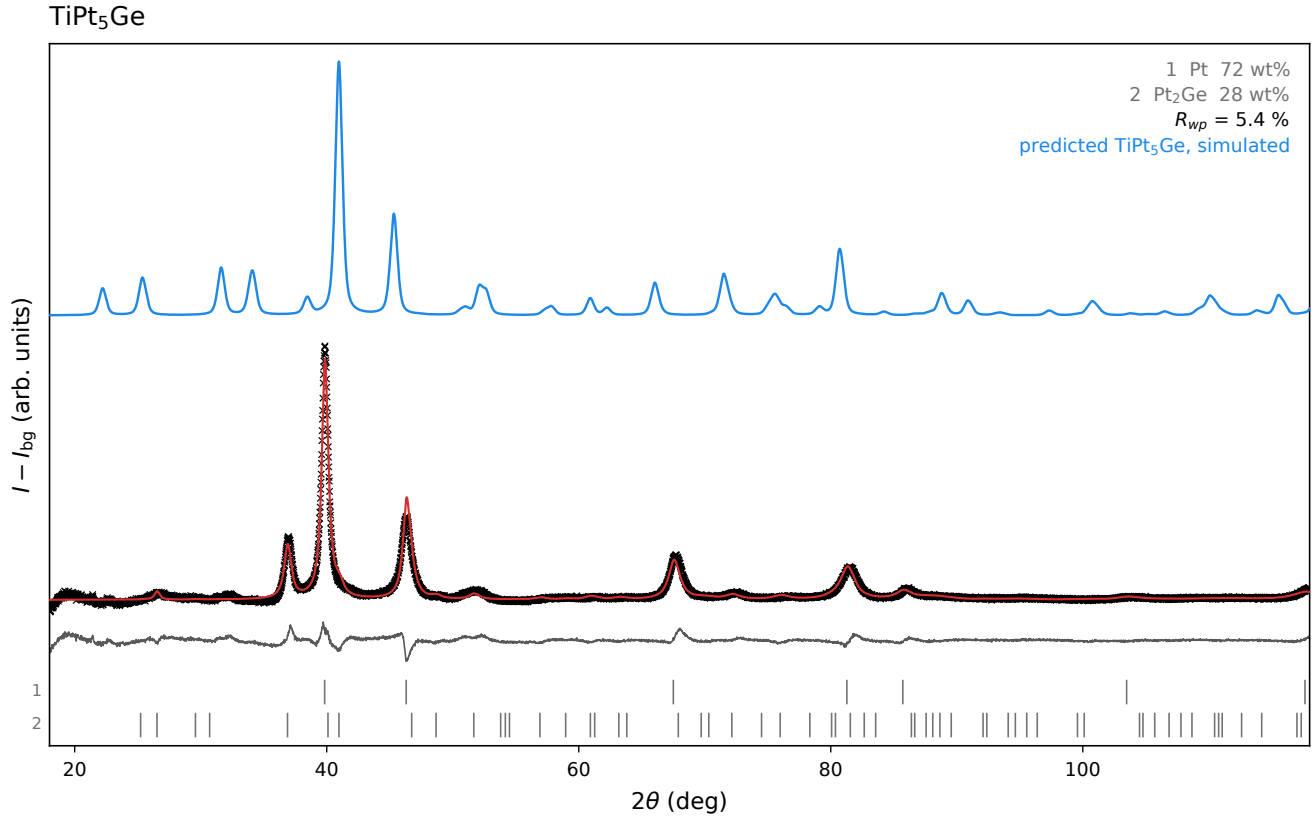


FIG. S23: $TiPt_5Ge$. Measured pattern with the background subtracted (black crosses), the best fit (red) and its residual (gray), with the Bragg positions of its phases below. Above them, offset upward, the pattern of the predicted structure simulated from its computed crystal structure, with line widths taken from the measured pattern (blue), scaled to the strongest observed line.

4. Ti_2Pt_8Si

The predicted structure does not match the pattern of Ti_2Pt_8Si. The best match we found has two phases in nearly equal amounts, a Pt-rich fcc-derived Ti–Pt phase, modeled with a $TiPt_2$-type cell, and a phase with the predicted $P4/mmm$ structure of $TiPt_5Si$. The line near 34.7° belongs only to this $TiPt_5Si$-type phase (Fig. S24). With the predicted Ti_2Pt_8Si structure in its place, that line is left unexplained and R_{wp} rises from 7.9% to 14.5%. The search thus found a predicted structure, that of $TiPt_5Si$ (Sec. S7 S7.1 4), in a sample whose nominal composition is different.

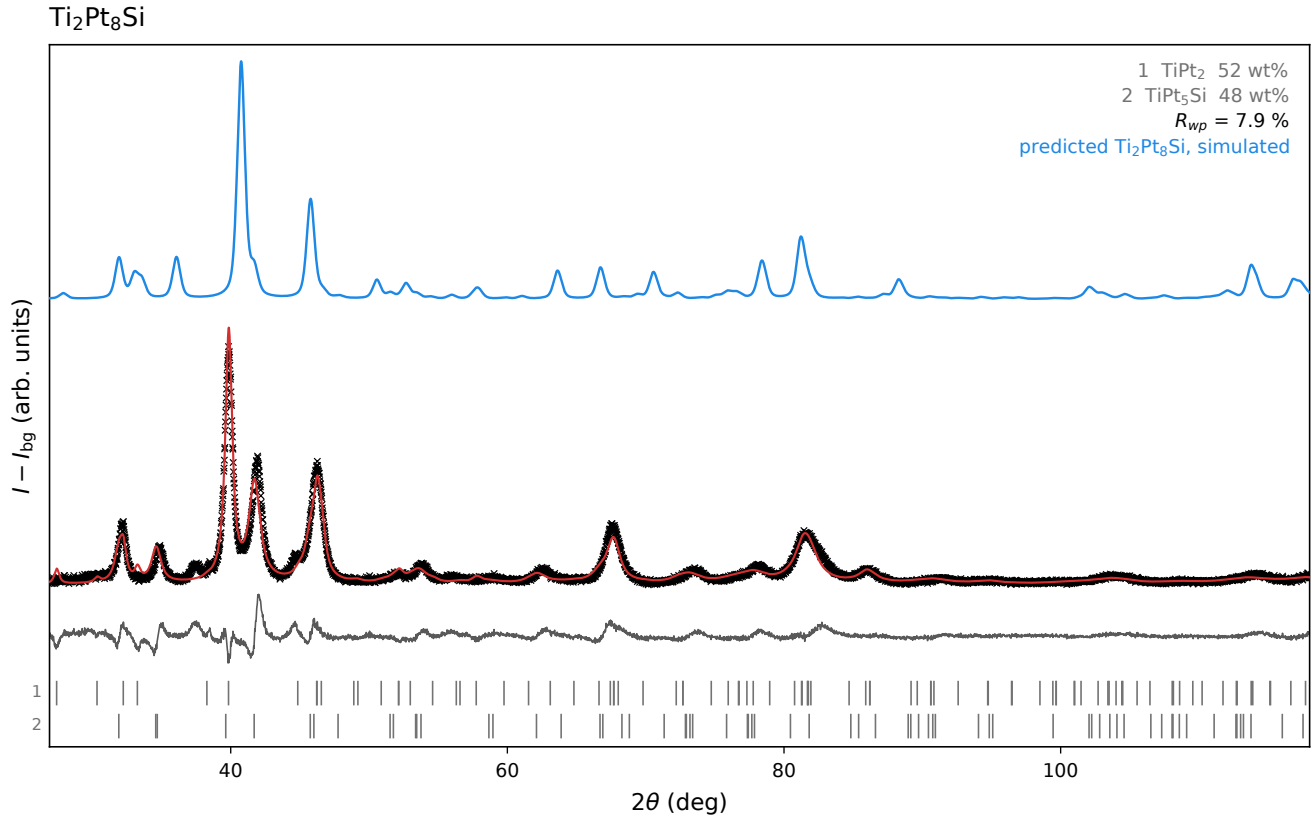


FIG. S24: Ti_2Pt_8Si. Measured pattern with the background subtracted (black crosses), the best fit (red) and its residual (gray), with the Bragg positions of its phases below. Above them, offset upward, the pattern of the predicted structure simulated from its computed crystal structure, with line widths taken from the measured pattern (blue), scaled to the strongest observed line. Sample displacement held at zero (Sec. S3).

5. $LaNiSi_3$

The predicted structure does not match the pattern of $LaNiSi_3$. Refined alone, the predicted $I4mm$ structure gives $R_{wp} = 41.3\%$ (Fig. S25). A mixture of known La–Ni silicides, mainly $La_3Ni_3Si_7$ and $LaNiSi_2$, with Si and probably some $LaNi_6Si_6$ in the structure of Pani et al. [11], matches the pattern well ($R_{wp} = 10.6\%$).

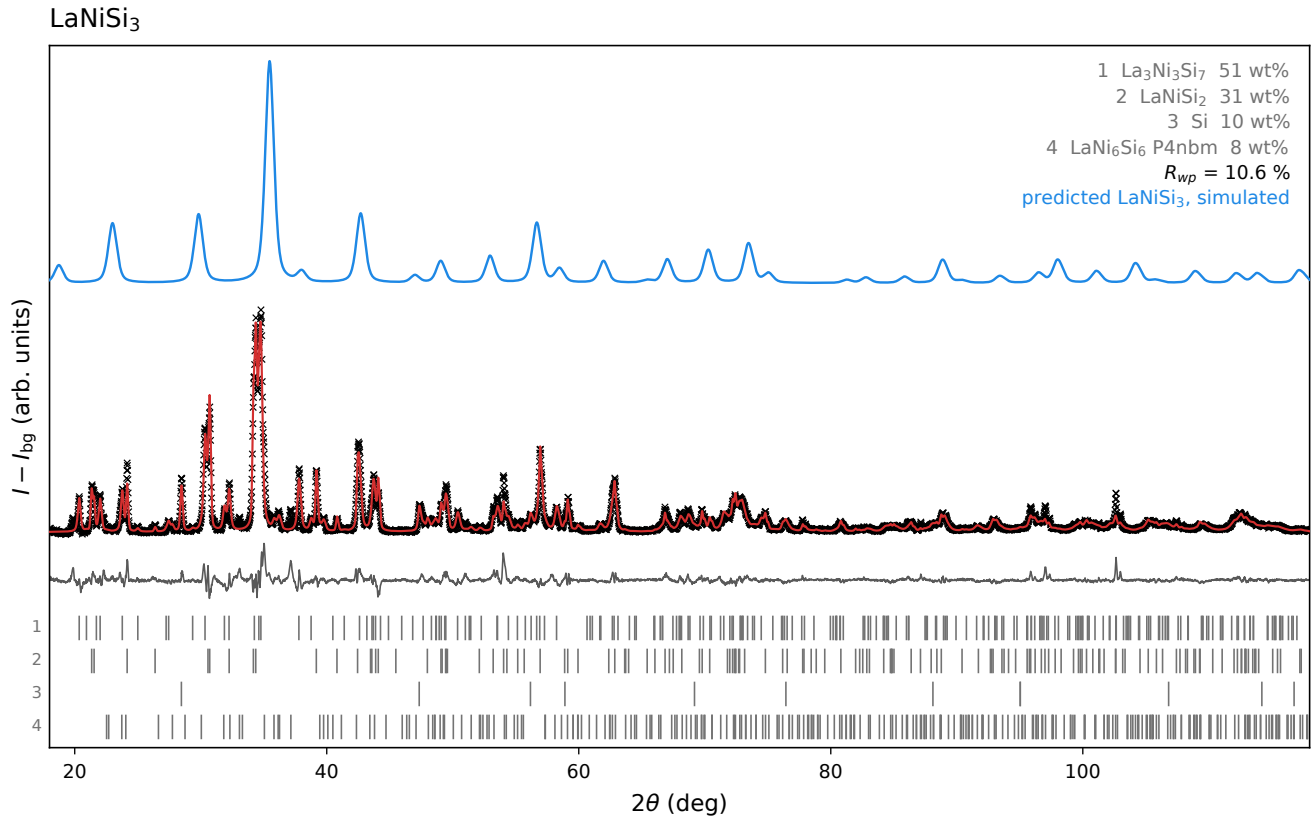


FIG. S25: $LaNiSi_3$. Measured pattern with the background subtracted (black crosses), the best fit (red) and its residual (gray), with the Bragg positions of its phases below. Above them, offset upward, the pattern of the predicted structure simulated from its computed crystal structure, with line widths taken from the measured pattern (blue), scaled to the strongest observed line.

6. $GdSn_6Co_4$

The predicted structure is not identified in the pattern of $GdSn_6Co_4$. Its strongest line coincides with lines of other phases, and at its next strongest lines, near 31.3, 41.7 and 43.0°, the data show no intensity beyond the fit. The best match we found combines five known phases and fits the pattern well (R_{wp} = 7.8%, Fig. S26). The two largest, a $GdCo_6Sn_6$-type phase and $Gd_3Co_4Sn_{13}$ (about 38 and 27 wt%), have several lines of their own and are clearly present. The three smaller ones, $GdCo_xSn_2$ (refined with the structure of Gil et al. [9]), CoSn and Sn, are less certain.

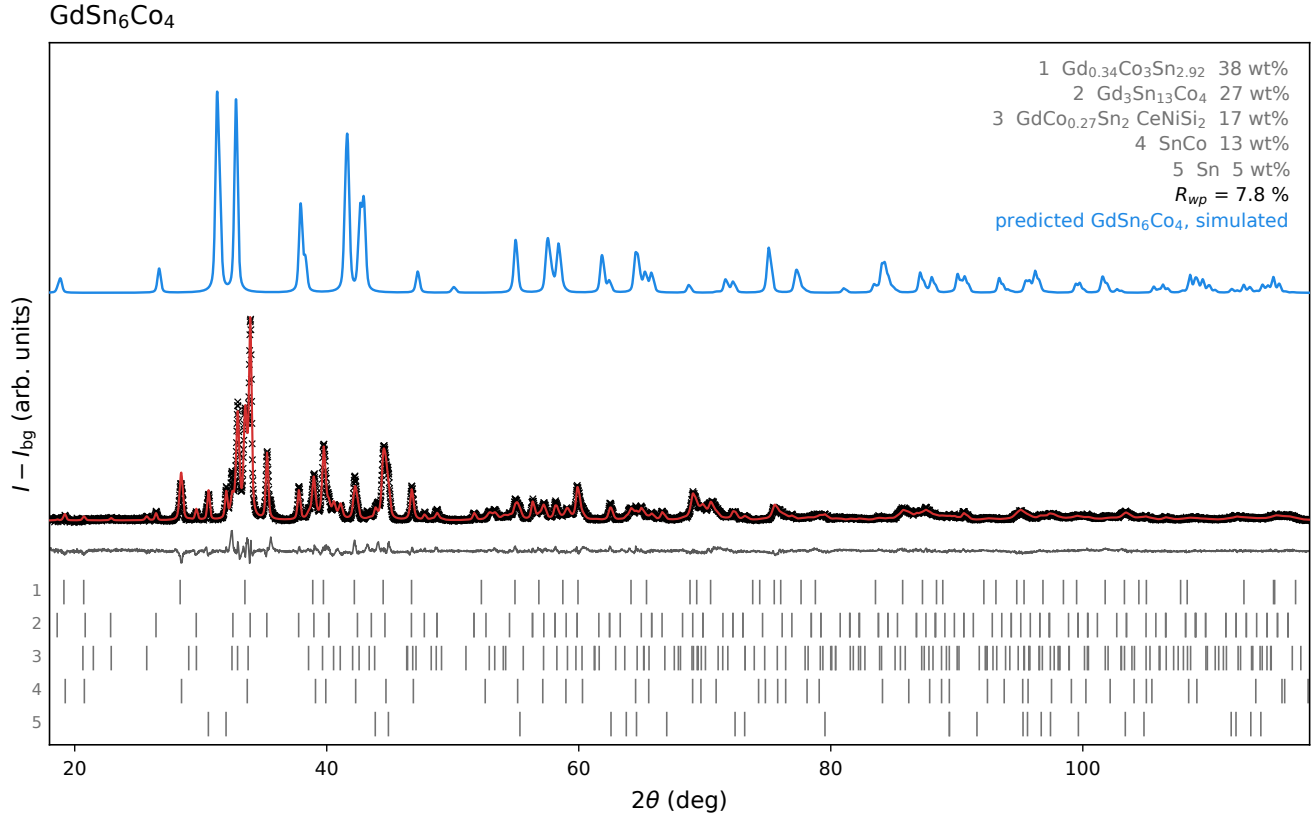


FIG. S26: $GdSn_6Co_4$. Measured pattern with the background subtracted (black crosses), the best fit (red) and its residual (gray), with the Bragg positions of its phases below. Above them, offset upward, the pattern of the predicted structure simulated from its computed crystal structure, with line widths taken from the measured pattern (blue), scaled to the strongest observed line.

7. $NbNi_2B_2$

The predicted structure does not match the pattern of $NbNi_2B_2$. Refined alone, the predicted $I4/mmm$ structure gives $R_{\rm wp} = 38.4\%$, and its strong (101) line at 28.40° falls where the data show only background (Fig. S27). A mixture of Ni_3B, NbB_2 and the τ-boride $Nb_2Ni_{21}B_6$ matches the pattern well ($R_{\rm wp} = 11.3\%$).

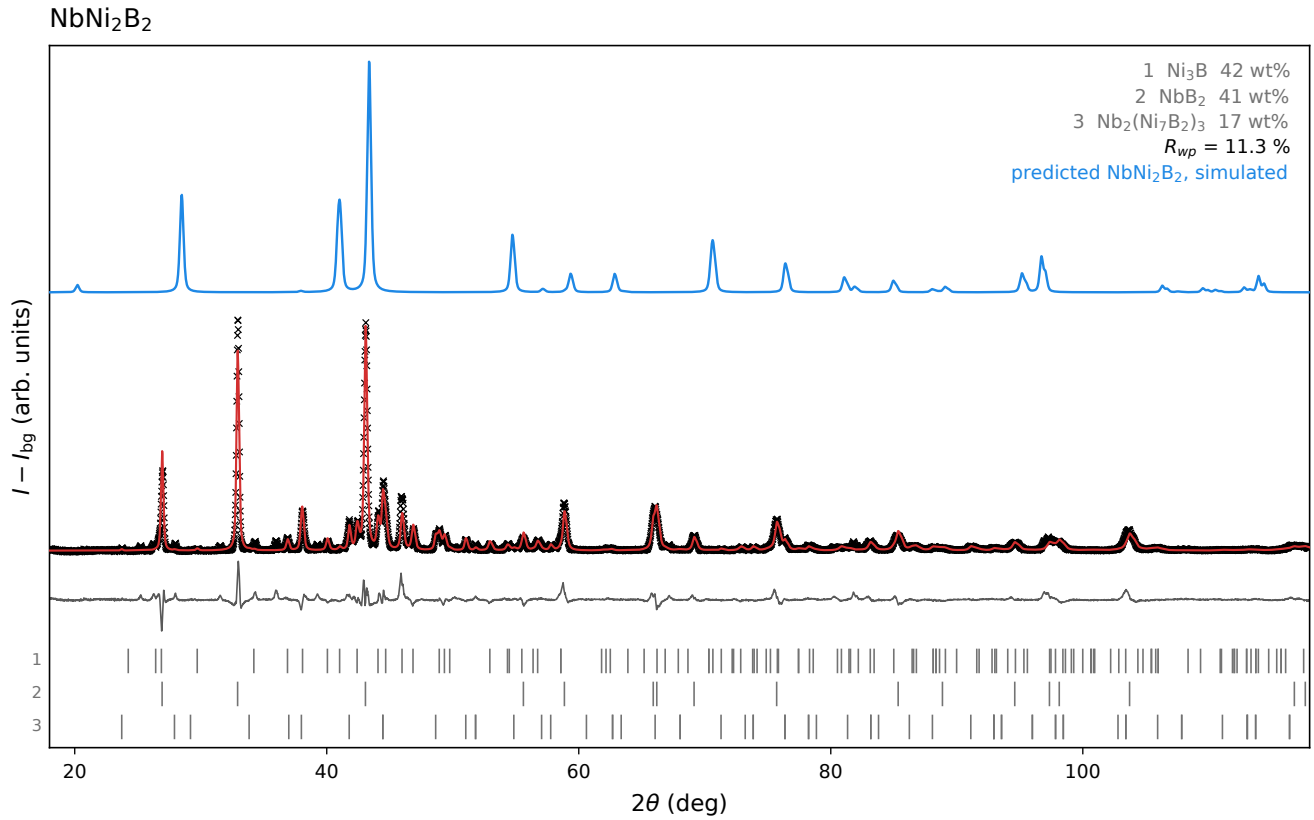


FIG. S27: $NbNi_2B_2$. Measured pattern with the background subtracted (black crosses), the best fit (red) and its residual (gray), with the Bragg positions of its phases below. Above them, offset upward, the pattern of the predicted structure simulated from its computed crystal structure, with line widths taken from the measured pattern (blue), scaled to the strongest observed line.

8. $GdFeGe_2$

The predicted structure does not match the pattern of $GdFeGe_2$. Refined alone, the predicted $P2_1/m$ structure gives $R_{wp} = 37.4\%$, and its isolated lines at 29.92 and 42.91° are absent (Fig. S28). A mixture of known Gd–Fe–Ge phases, mainly $GdFe_{0.37}Ge_2$ and $GdFe_2Ge_2$, matches the pattern well ($R_{wp} = 7.3\%$).

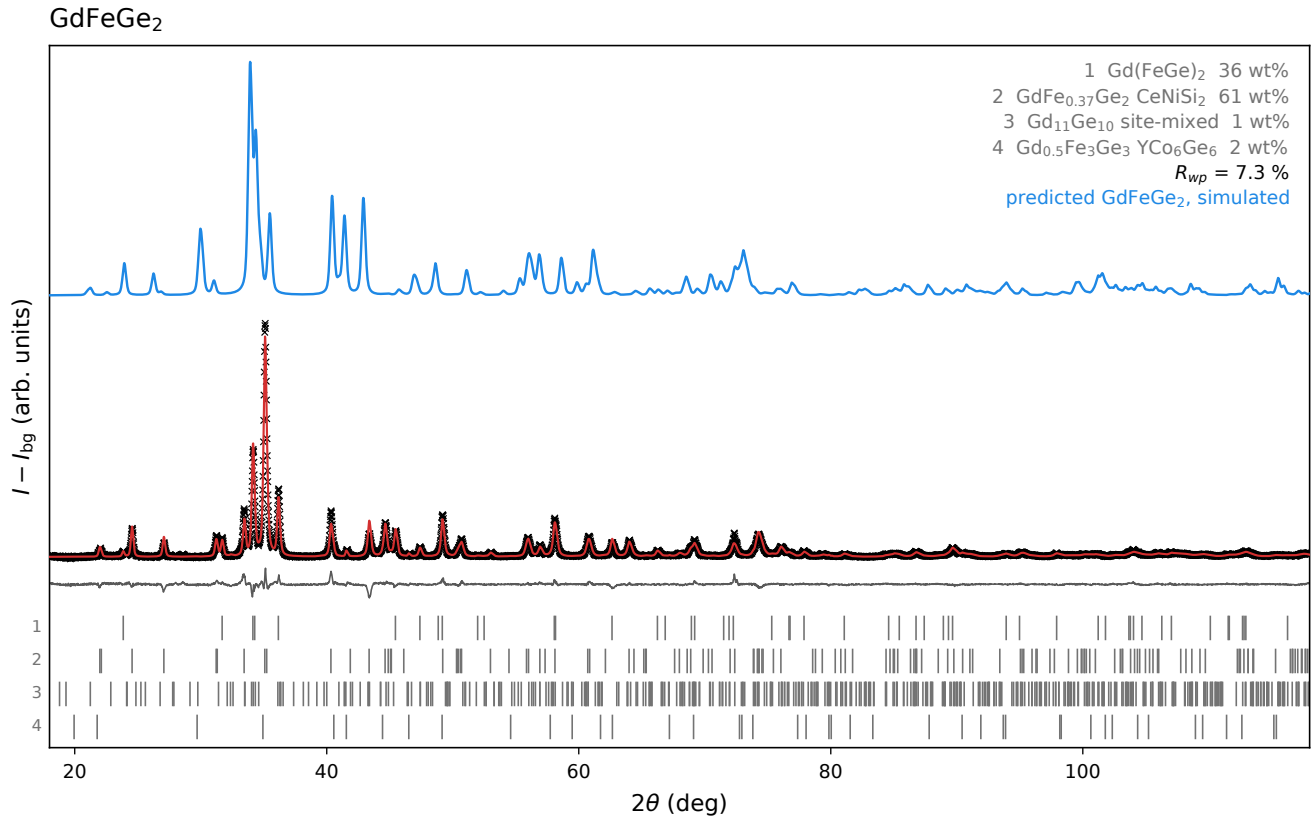


FIG. S28: $GdFeGe_2$. Measured pattern with the background subtracted (black crosses), the best fit (red) and its residual (gray), with the Bragg positions of its phases below. Above them, offset upward, the pattern of the predicted structure simulated from its computed crystal structure, with line widths taken from the measured pattern (blue), scaled to the strongest observed line.

9. $Gd_2Fe_3Ge_5$

The predicted structure does not match the pattern of $Gd_2Fe_3Ge_5$. Refined alone, the predicted $C2/c$ structure gives $R_{wp} = 38.5\%$ (Fig. S29). A mixture of known Gd–Fe–Ge phases, mainly $GdFe_{0.37}Ge_2$, $GdFe_2Ge_2$ and $Gd_{0.5}Fe_3Ge_3$, with likely a little Ge, matches the pattern well ($R_{wp} = 9.4\%$).

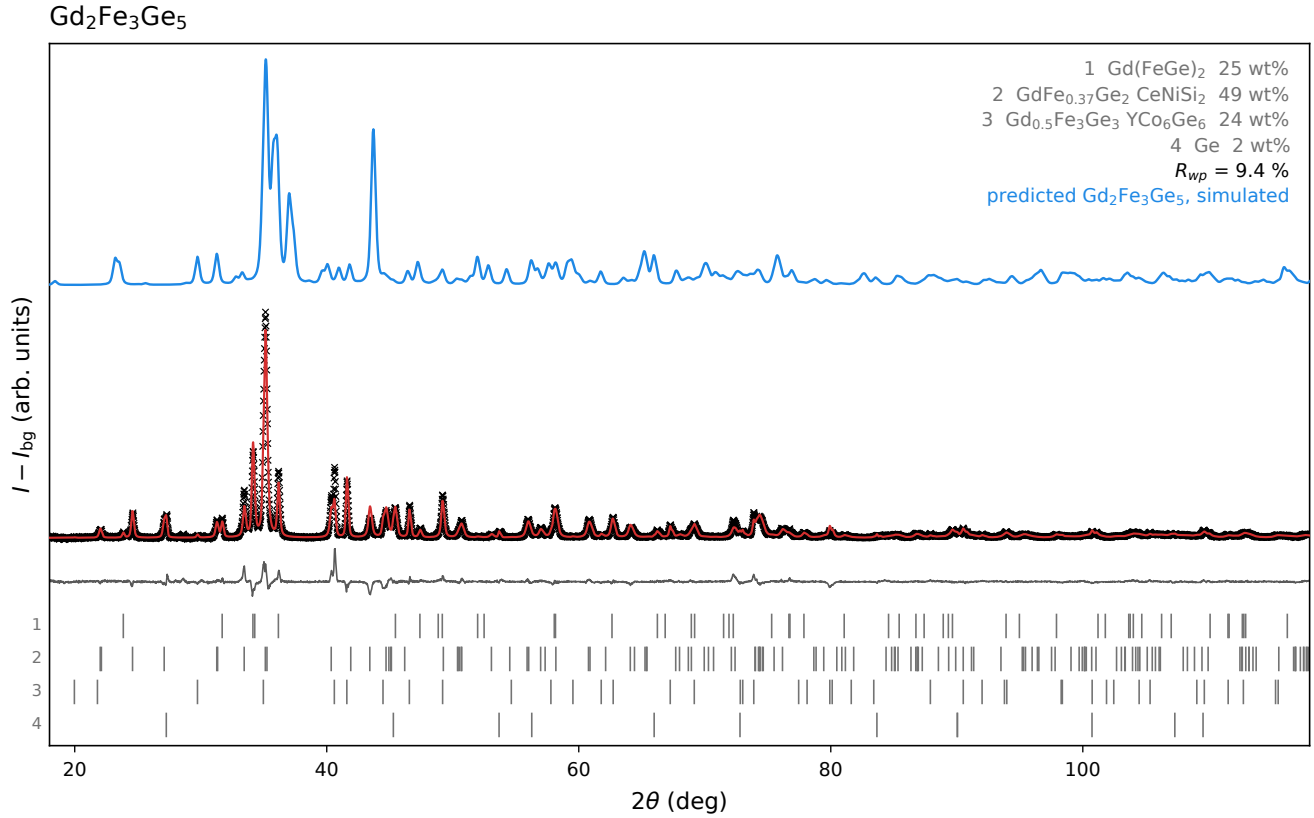


FIG. S29: $Gd_2Fe_3Ge_5$. Measured pattern with the background subtracted (black crosses), the best fit (red) and its residual (gray), with the Bragg positions of its phases below. Above them, offset upward, the pattern of the predicted structure simulated from its computed crystal structure, with line widths taken from the measured pattern (blue), scaled to the strongest observed line.

10. $GdFeGe_3$

The predicted structure does not match the pattern of $GdFeGe_3$. Refined alone, the predicted $I4mm$ structure gives $R_{wp} = 39.9\%$, and its isolated (103) and (200) lines, at 34.43 and 42.31°, are absent (Fig. S30). A mixture of $GdFe_{0.37}Ge_2$, $Gd_{0.5}Fe_3Ge_3$ and Ge matches the pattern well ($R_{wp} = 7.5\%$). A weak line at 27.98° is not explained.

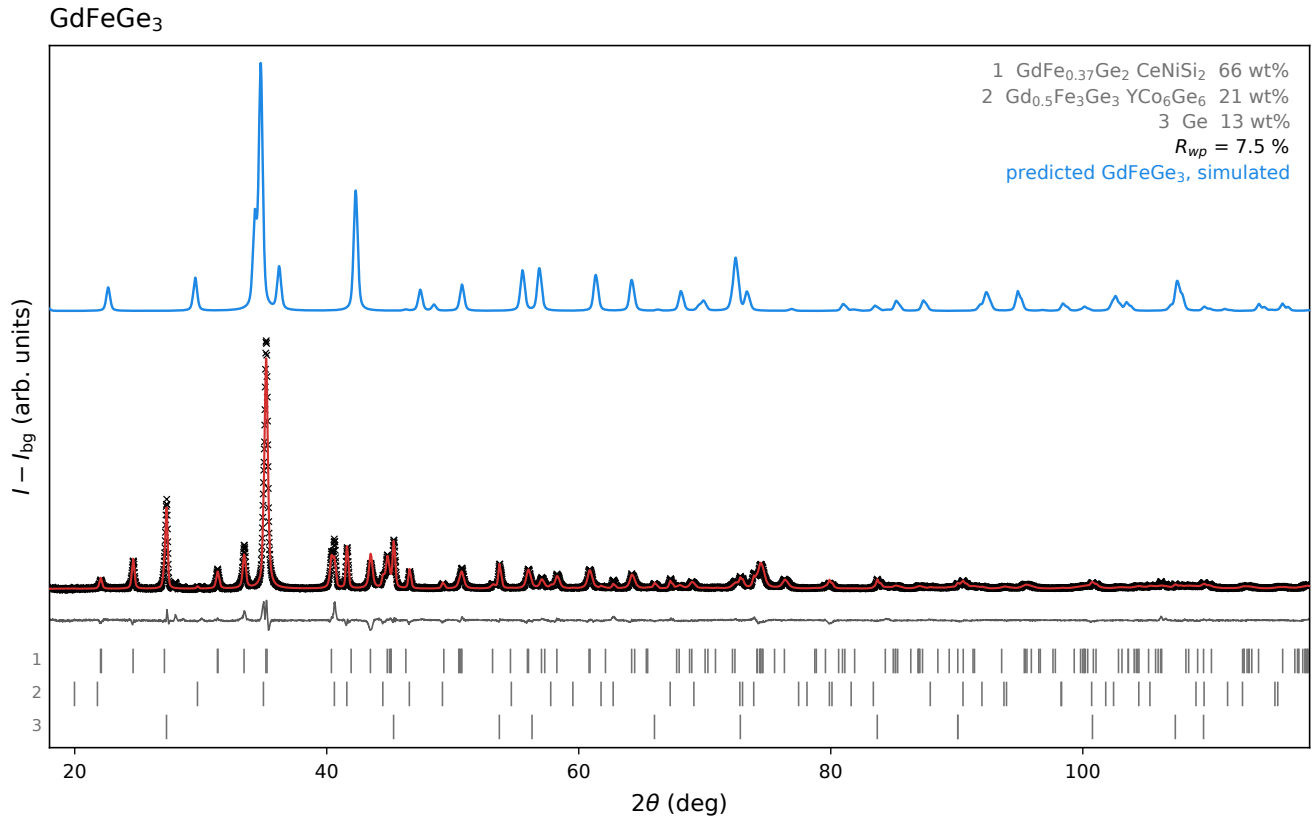


FIG. S30: $GdFeGe_3$. Measured pattern with the background subtracted (black crosses), the best fit (red) and its residual (gray), with the Bragg positions of its phases below. Above them, offset upward, the pattern of the predicted structure simulated from its computed crystal structure, with line widths taken from the measured pattern (blue), scaled to the strongest observed line.

11. $LaRuC_2$

The predicted structure of $LaRuC_2$ does not match the pattern of the button with the smallest mass loss (Fig. S31). Refined alone it fits far worse than our best fit ($R_{wp} = 40.6\%$ against 7.5%). The best fit uses Ru metal and $La(OH)_3$. $La(OH)_3$ was not expected, and its presence suggests that the sample oxidized. An earlier button, which lost more mass during melting, gave a different pattern that we could not describe, but the predicted structure did not form there either.

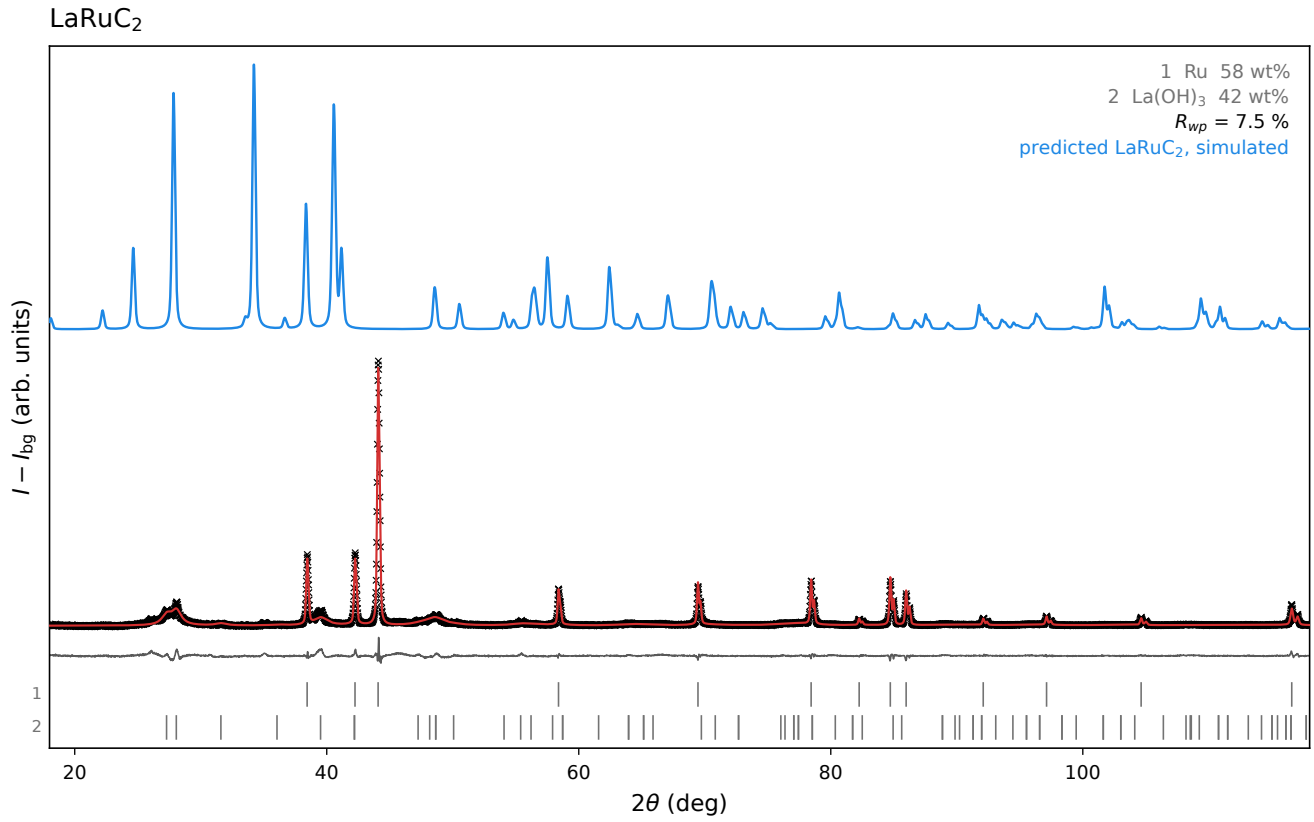


FIG. S31: $LaRuC_2$. Measured pattern with the background subtracted (black crosses), the best fit (red) and its residual (gray), with the Bragg positions of its phases below. Above them, offset upward, the pattern of the predicted structure simulated from its computed crystal structure, with line widths taken from the measured pattern (blue), scaled to the strongest observed line.

S7.6. Samples whose phases are unresolved, without the on-hull structure

1. Pd_6Cu_5Ge

The predicted *Cmmm* structure of Pd_6Cu_5Ge does not match the pattern (Fig. S32). Refined alone it fits much worse than our best fit (R_{wp} = 14.4% against 8.7%), and its superlattice lines near 29.5°, which would stand about six times the noise above the background, are absent. The best fit uses a broad fcc solid solution and a small amount of a bcc phase. Because the lines are so broad, this fit does not establish the composition or order of the fcc phase, so we quote no weight fractions and leave what formed unresolved.

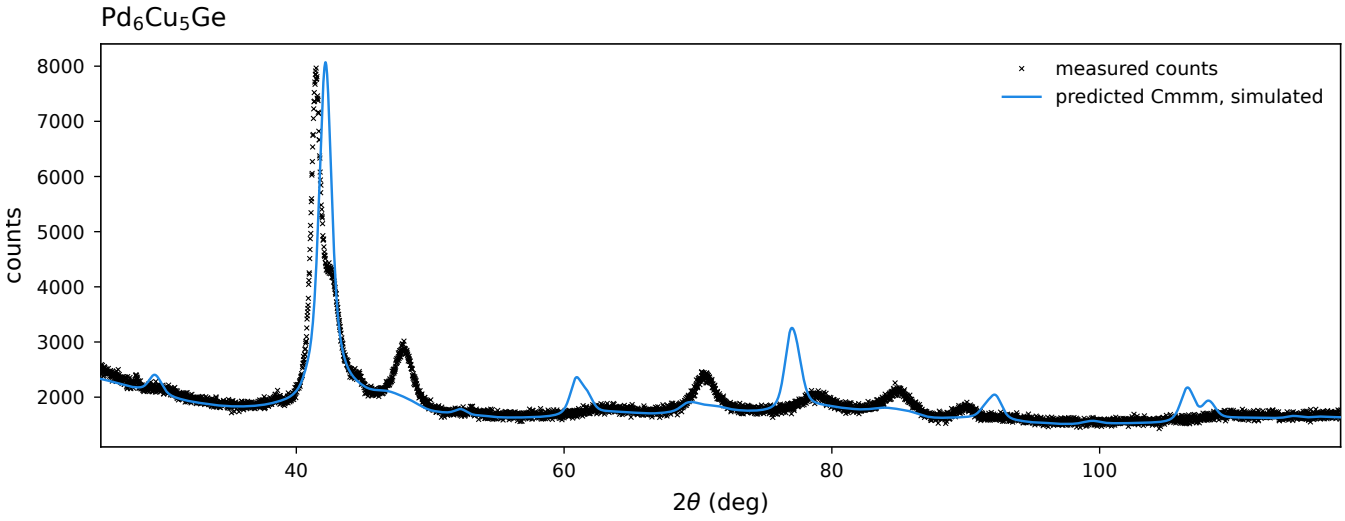


FIG. S32: Pd_6Cu_5Ge, powder. Measured counts (black crosses) and the pattern of the predicted structure simulated from its computed crystal structure, with line widths taken from the measured pattern (blue), scaled to the strongest observed line and added to a smooth background estimate. No fit is shown because we have no satisfactory description of the pattern.

2. La_2PdAu_2

The pattern of La_2PdAu_2 is not well explained. No combination of phases we tried fits it, and the best one misses about half the intensity of the strongest line, at 32.54° (Fig. S33). The predicted $C2/m$ structure is not present, because its two strongest lines, at 31.13 and 37.32°, show no intensity above the background. What formed is not identified.

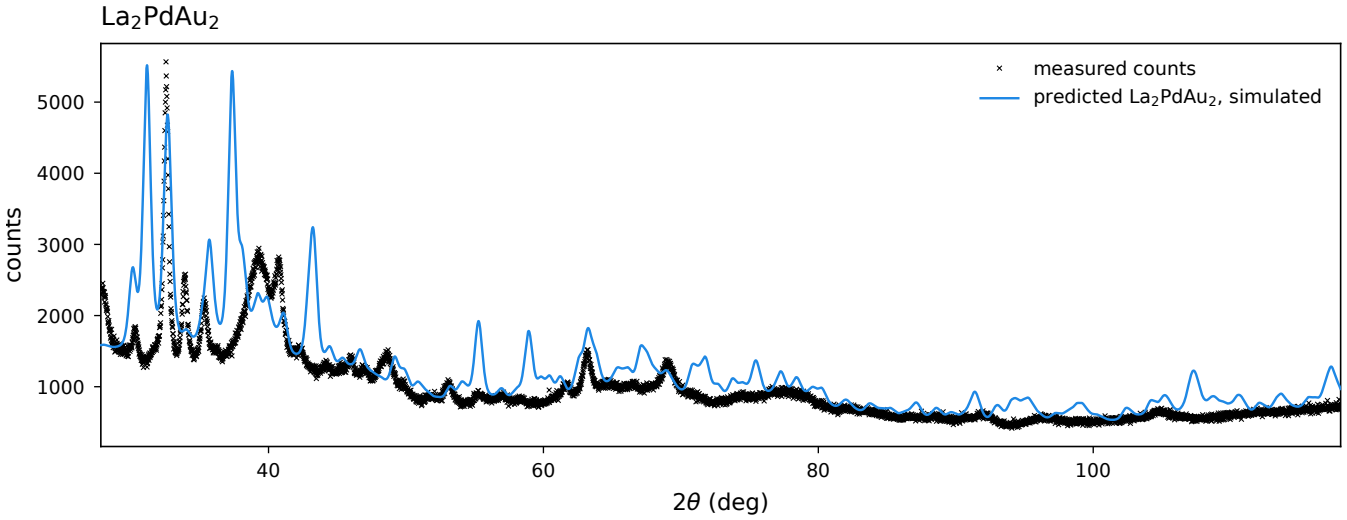


FIG. S33: La_2PdAu_2. Measured counts (black crosses) and the pattern of the predicted structure simulated from its computed crystal structure, with line widths taken from the measured pattern (blue), scaled to the strongest observed line and added to a smooth background estimate. No fit is shown because we have no satisfactory description of the pattern.

3. Ti_2Pt_8Ge

The predicted $P4/mmm$ structure of Ti_2Pt_8Ge does not match the pattern, which was measured on a saw-cut face of the as-cast button (Fig. S34). The strongest observed lines index on a nearly cubic, Pt-rich fcc-type cell, while the tetragonal cell of the prediction places lines at 40.4 and 49.8°, where the data show none. We have no satisfactory description of the whole pattern, so what formed is not identified.

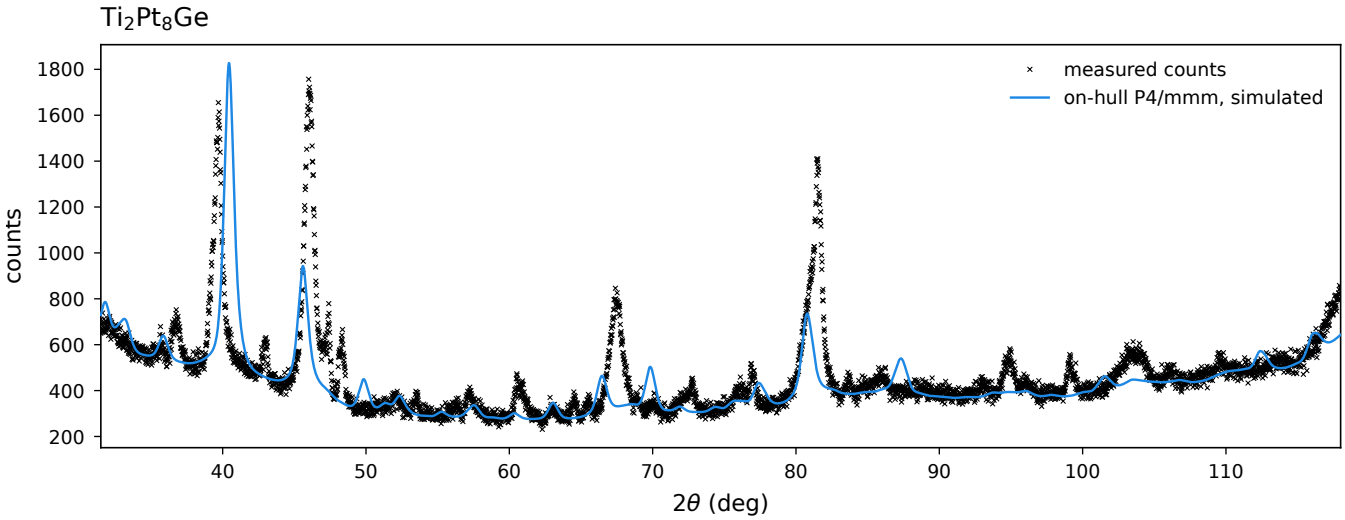


FIG. S34: Ti_2Pt_8Ge, as-cast, measured on a saw-cut face. Measured counts (black crosses) and the pattern of the predicted structure simulated from its computed crystal structure, with line widths taken from the measured pattern (blue), scaled to the strongest observed line and added to a smooth background estimate. No fit is shown because we have no satisfactory description of the pattern.

4. $ScPdCu_4$

The patterns of $ScPdCu_4$ show only broad features, without well-defined lines (Fig. S35). The three buttons we measured gave similar patterns (Table S1). The pattern computed from the predicted structure does not match them, and we could not identify the phases that formed.

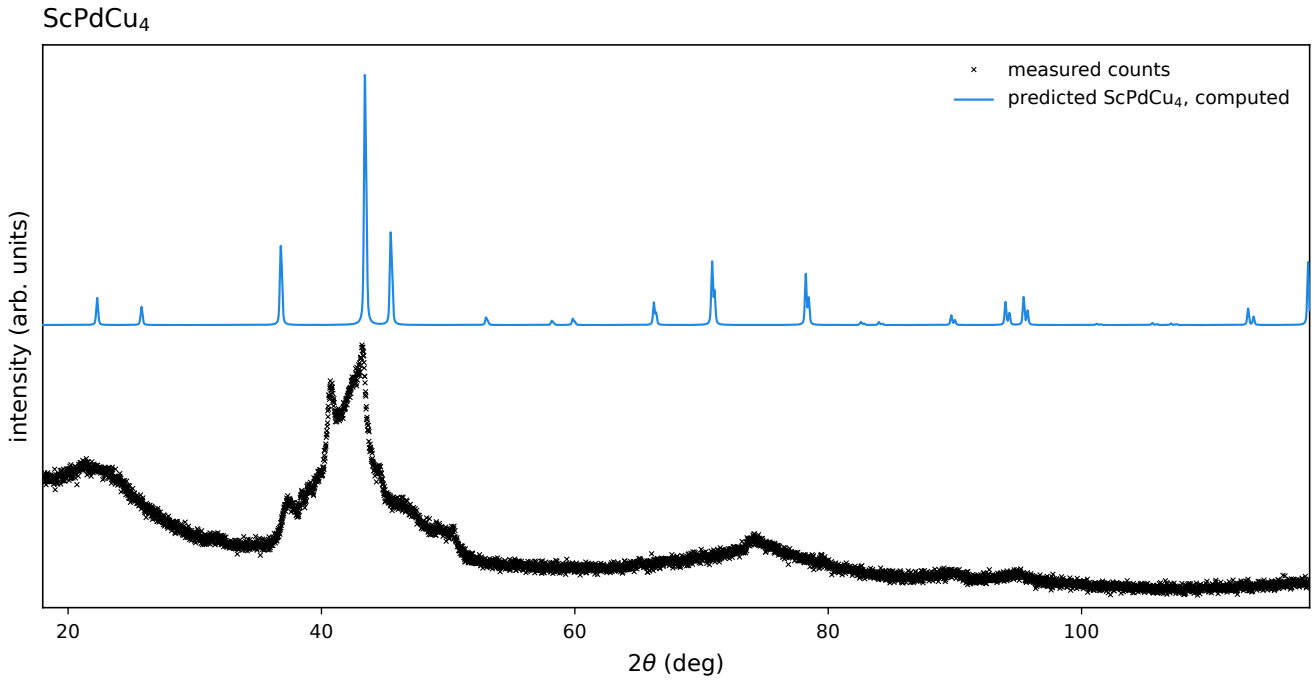


FIG. S35: $ScPdCu_4$. Measured counts (black crosses) and, above them, the pattern computed from the predicted structure with a line width of 0.15° and no background (blue). No fit is shown because we have no satisfactory description of the pattern.

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